

Currency Value*

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Abstract

We assess the properties of currency value strategies based on real exchange rates in a cross-sectional portfolio setting. We find that real exchange rates predict currency excess returns, but in a way that is inconsistent with the notion of currency value because a high valuation level forecasts a relative appreciation of the foreign currency going forward. However, adjusting real exchange rates for two key country-specific fundamentals – productivity and the quality of export goods – generates a measure of currency value which still predicts currency excess returns and is consistent with the spot exchange rate reverting toward fundamental value.

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1 Introduction

The real exchange rate (RER) is a key measure of currency valuation, used widely by both policy makers and market participants.¹ The general idea is that misaligned real exchange rates should adjust to their fundamental value over time, i.e., that exchange rates which are overvalued relative to fundamental values will depreciate and vice versa, at least over the long-run. The existence and nature of this adjustment mechanism plays a key role in international finance models, for policy makers and investors, but is far from being well understood. This paper sheds new light on the properties of currency value by addressing the following questions in the context of currency value investment strategies: Do measures of currency valuation predict nominal exchange rate changes and currency excess returns? If so, is this predictability consistent with the notion of currency value such that spot exchange rates tend to correct disequilibria? Do simple purchasing power parity (PPP) calculations provide accurate measures of currency value or, alternatively, need to be adjusted for country-specific fundamentals?

We tackle these questions in a multi-currency portfolio setup, providing us with a straightforward assessment of the economic significance of the link between real exchange rates, spot exchange rate changes, and risk premia.² Importantly, we find that simple RER levels do not do a good job of capturing “currency value” in its original sense but that it is necessary to adjust the RER for structural cross-country differences in key macro fundamentals. These adjusted real exchange rates provide a much cleaner measure of currency misalignment.

Specifically, we show that countries with a weak RER against the U.S. dollar (that is, the

¹In the financial industry, for example, exchange-traded funds that trade currencies on the basis of a real exchange rate value signal are quite common (e.g., the Deutsche Bank Currency Valuation Index). In the policy making community, there is a long history of using models of real exchange rate equilibrium to gauge exchange rate misalignments in central banks and international organizations, notably at the IMF.

²For earlier approaches based on forming currency portfolios see, e.g., Lustig and Verdelhan (2007); Lustig, Roussanov, and Verdelhan (2011); Burnside, Eichenbaum, Kleshchelski, and Rebelo (2011); Menkhoff, Sarno, Schmeling, and Schrimpf (2012a). These papers typically investigate returns to carry trades or closely related portfolios. The focus in this paper is on real exchange rates and related valuation measures which have received much less attention in the recent finance literature – the exceptions being Asness, Moskowitz, and Pedersen (2012); Barroso and Santa-Clara (2014); Kroencke, Schindler, and Schrimpf (2014); Pojarliev and Levich (2010). Heyerdahl-Larsen (2014) calibrates a model with deep habits and consumption bias to match the cross-sectional evidence on portfolios sorted on both carry and value. Balduzzi and Chiang (2014) investigate time-series return predictability of currency returns by real exchange rates, and Engel (2011) investigates the link between RER levels and (real) interest rate differentials in the time series dimension.

currency is cheap in real terms compared to the dollar) have higher currency excess returns going forward than countries with a strong RER. A natural interpretation might be that currencies with a low valuation level appreciate in the future and thus deliver high excess returns as suggested by standard PPP reasoning. Contrary to this intuition, however, we find that low valuation levels actually predict a significant relative *depreciation* of the currency, which is just the opposite of the intuition for currency value strategies. In fact, the high excess returns to currencies with low valuations are driven by the carry component (higher interest rates). This means that, while RER levels contain predictive information for currency excess returns in the cross section of currencies, this predictability is not consistent with the notion of currency value as the returns are generated by buying (selling) high (low) interest rate currencies rather than predicting future spot returns.

We suggest an explanation for this new result based on persistent structural differences across countries: Some countries feature persistently high valuation levels and strong real exchange rates; currencies of such countries exhibit low risk premia and tend to experience further nominal appreciations going forward. Other countries, by contrast, have persistently low valuation levels, their nominal exchange rates tend to depreciate on average, and investors demand a positive risk premium for holding such currencies. In line with this persistence, the current RER level carries almost the same information for future returns as RER levels several years back in time. This finding points towards important persistent (or *static*) components in exchange rate predictability by the RER.

To provide a better understanding of these risk premium components we investigate their potential economic underpinnings. Our empirical analysis suggests that observable structural differences across countries can indeed account for large parts of the predictability pattern. We adjust real exchange rates across countries by considering two key variables that account for persistent differences in price levels: (a) Harrod-Balassa-Samuelson (HBS) effects, and (b) the quality of a country's exports. In this way, we remove from the RER the impact of country characteristics that can prevent goods market arbitrage from restoring PPP, thus delivering a more accurate measure of "fair value." We find that the HBS effect (captured by differences in real per capita GDP) and differences in export quality across countries explain a large share of the cross-country variation in real exchange rates. Accounting for such

structural cross-country differences eliminates the puzzling link between real exchange rates and future currency appreciations and depreciations. In fact, using the HBS- and quality-adjusted measures of real exchange rates, we find that an undervaluation relative to fair value predicts a high excess return going forward, which is almost fully driven by an appreciation of the currency, in line with standard intuition for currency value strategies.³

To further understand the predictive power of real exchange rates and underlying drivers of currency risk premia, we draw on present-value formulations of exchange rates (see, e.g., Froot and Ramadorai, 2005; Engel and West, 2005, 2006). We find that output gaps account for the lion's share of risk premium predictability whereas inflation and interest rate differentials matter most for exchange rate predictability by the RER. This finding squares well with standard theory and earlier results where measures of the business cycle (output gaps) matter for time-variation in risk premia (see, e.g., Lustig, Roussanov, and Verdelhan, 2014) and where nominal fundamentals (interest rates, inflation) matter for exchange rates (e.g., Engel and West, 2005; Sarno and Schmeling, 2014). We thus identify a link between macro fundamentals and currency risk premia in the cross-section of countries which operates via the connection between macro fundamentals and real exchange rates.

Finally, we also investigate the information content of long-run (five-year) RER changes as a measure of currency value as is done in the recent finance literature (e.g. Asness, Moskowitz, and Pedersen, 2012; Kroencke, Schindler, and Schrimpf, 2014; Barroso and Santa-Clara, 2014). Five-year changes in RERs illustrate a somewhat different predictive behavior for excess returns. While RER levels have greater predictive content for exchange rates themselves, five year changes in real exchange rates have greater predictive power for excess returns but much less predictive power for exchange rate changes. This finding can be understood in light of our empirical results that exchange rate predictability is mostly driven by fairly persistent differences in country characteristics. Using five-year RER changes effectively mitigates the impact of these persistent differences and isolates a value measure that is more closely tied to business cycle frequency movements in countries' valuations. That said, adjusting five-year RER changes for HBS and quality effects still improves forecast power, and the adjusted value measure strongly predicts both currency excess returns and exchange rate changes.

³On a related note, see Jorda and Taylor (2012) on how fundamental exchange rate values matter for carry trade returns, and Chong, Jorda, and Taylor (2012) on how HBS effects matter for long-run PPP.

Overall, our contributions to the literature are threefold. First, we provide an extensive analysis of currency value strategies based on real exchange rates in a cross-sectional portfolio setting.⁴ This multi-currency investment approach provides an intuitive measure of the economic value of signals based on real exchange rates and allows us to pin down the basic properties of the returns from value strategies. Second, while we are not the first to show that long-term changes in real exchange rates forecast currency returns in a portfolio setting, we shed light on the nature of the macroeconomic information that drives this forecasting power, which we can attribute clearly to country-specific fundamentals. Third, while we find that simple PPP calculations do not provide an accurate measure of currency value in the sense that spot exchange rates do not tend to converge to PPP in our broad cross-section, we show how adjusting real exchange rates for productivity differentials and the quality of export goods delivers an accurate measure of currency value that displays strong predictive power for both currency returns and currency excess returns.

Our results are obtained in an out-of-sample portfolio setting, which is important as the vast majority of papers in the literature either do not consider out-of-sample forecasting at all when analyzing real exchange rate models or rely on purely statistical performance measures derived from time-series analysis of a limited number of currency pairs (see, e.g., Rogoff, 1996; Taylor and Taylor, 2004). Time-series analysis is affected by trends in exchange rates, most notably the U.S. dollar. Our portfolio procedure, by contrast, studies exchange rate predictability in dollar-neutral long-short portfolios. Moreover, we are the first to explore the role of differences in the quality of exports across countries for real exchange rate convergence to equilibrium, which is important because the notion of arbitrage in international goods markets which underlies currency value assumes that real exchange rate convergence across countries occurs for identical goods.

Taken together, these results have implications for our general understanding of value in FX markets and provide a novel investment strategy which should be useful for asset managers interested in diversifying away from conventional strategies such as carry and momentum.

The paper proceeds as follows. To set the stage, Section 2 lays out how we decompose

⁴See Melvin and Shand (2013) on the relevance of cross-sectional forecasts for actual implementations of currency strategies.

the RER into various economic drivers. Section 3 presents the data. In Section 4 we show the predictive power of real exchange rates in a multi-currency investment strategy setup. Then, in Section 5, we look at this result from various angles to shed light on how risk premia and the concept of currency value are linked. Robustness analyses and further results are discussed in Section 6. Section 7 concludes.

2 Decomposing the real exchange rate

In this section, we lay out the main relationships between RER, economic fundamentals, and other measures of currency value we are interested in. This sets the stage for the subsequent empirical analysis, where we first look at the predictive content of the RER and then dissect its information content into various underlying drivers to gain a better understanding of currency risk premia.

The RER, Q is the most common measure of currency valuation and we work with the following definition

$$Q_t = \frac{P_t}{S_t P_t^*}, \quad (1)$$

where S denotes the exchange rate change (USD per unit of foreign currency), P denotes the U.S. price level, and P^* denotes the foreign price level. The price levels are measured by the consumer price index, the producer price index, or the GDP deflator, and each of them contains a mix of tradable and non-tradable goods. Real exchange rates that differ from unity thus capture the deviation of a currency's value from PPP (see, e.g., Rogoff, 1996; Taylor and Taylor, 2004). The scaling here is such that a higher RER (Q) means a stronger dollar and, consequently, a *lower* valuation level of the foreign currency. The log real exchange rate is $q_t = p_t - p_t^* - s_t$, where lower-case letters denote logs of variables. Note that the nominal spot exchange rate, S , is defined in the opposite way, i.e., a higher value of S means that the foreign currency is more expensive (stronger) in nominal terms. We do so to simplify the presentation of results in the subsequent empirical analysis. We also discuss this issue in more detail below.

The definition of Q given above is derived from the law of one price applied to individual goods prices that are perfectly substitutable and internationally tradable. Relaxing the as-

sumption that all goods in the price indices P and P^* are tradable gives rise to HBS effects in that more productive economies tend to experience stronger real exchange rates. Also, relaxing the assumption of perfect substitutability – for example due to differences in quality of a country’s traded goods – introduces a wedge in prices such that countries producing higher-quality goods experience stronger real exchange rates. We take into account both HBS effects and export quality later in the paper when we construct our proposed measures of currency value.

2.1 A mechanical real exchange rate decomposition

As our first decomposition we split up the current RER level into the lagged RER level and the change in the level. This decomposition draws on recent work by Gerakos and Linnainmaa (2014) and allows us to gauge the persistence in the predictive power in RER levels. While Gerakos and Linnainmaa (2014) study the information content of equity valuation measures (book-to-market), their decomposition and main idea are directly applicable to currencies. Specifically, we decompose the log RER today, q_t , into RER five years ago (q_{t-5y}) plus the change in RER ($\Delta q_{t-5y;t}$) from five years ago until time t

$$q_t = \underbrace{(p_{t-5y} - p_{t-5y}^* - s_{t-5y})}_{\text{Past level } q_{t-5y}} + \underbrace{(\pi_{t-5y;t} - \pi_{t-5y;t}^* - \Delta s_{t-5y;t})}_{\text{Change until today } \Delta q_{t-5y;t}}, \quad (2)$$

where π denotes the log CPI inflation rate. With this timing of five years, the change in q is identical to value measures used elsewhere in the finance literature (see, e.g., Asness, Moskowitz, and Pedersen, 2012, for a recent example which uses five-year RER changes as their signal for currency value).

The five-year change in the RER can then be further decomposed into the five-year change in log spot nominal exchange rate and five-year log inflation differential (against the U.S.). Hence, a first simple decomposition is to split up the log RER into (i) lagged log RER levels, (ii) the five-year change in the log spot exchange rate, and (iii) the differential in the five-year log foreign and domestic inflation rates.

Regardless of its simplicity, this decomposition allows us to examine two important questions. First, does any information in real exchange rates for future excess returns and spot

rate changes stem from lagged RER levels or from changes in real exchange rates? It is particularly interesting to disentangle these two channels in light of the recent work of Hassan and Mano (2015) who show that currency premia are largely static in nature and driven by persistent country characteristics instead of time-series variation of these features. This question can be easily addressed within this simple framework. Second, it is interesting to disentangle whether predictability in RER changes stems from lagged nominal exchange rate changes or inflation differentials. Since RER changes are viewed as a measure of currency valuation in the literature, this analysis is relevant to gain a better understanding of the mechanism behind its predictive content, which should stem from spot rate changes to be consistent with the standard notion of currency value.

2.2 Decompositions based on “fair value”

We also draw on insights from the literature on currency fair value models (see Cenedese and Stolper, 2012, for a survey of this literature) to provide a better understanding of the information contained in the RER and its relation to “currency value”. More specifically, we rely on what is known in the literature as the “Penn effect,” i.e., the tendency of high income and productivity countries to have persistently higher price levels than low income and productivity countries. This difference is typically explained by a HBS effect, that is, productivity differentials between tradable and non-tradable goods (Rogoff, 1996).

While this effect can explain why PPP may not hold empirically, there are additional factors that may contribute to persistent differences in price and RER levels across countries. One potential dimension that directly relates to goods market arbitrage and PPP deviations is the quality of export goods, which has been shown to be related also to income, trade, and growth patterns across countries (see, e.g., Hummels and Klenow, 2005; Henn, Papageorgiou, and Spatafora, 2013). Intuitively, products that differ in terms of their quality are not perfect substitutes and goods market arbitrage will not eliminate price differences between, say, high and low quality cars or high and low quality wine. Hence, we will add the effect of product quality differentiation to the standard Penn effect in the empirical analysis below.

Framework. To motivate our fair value decomposition and the two effects discussed above, we draw on the standard textbook setup (see, e.g., Obstfeld and Rogoff, 1996) in which countries only produce two goods, a tradable (T) and a non-tradable (N) good. Capital is mobile, both across countries and across sectors within countries, whereas labor is mobile across sectors but not across countries.

A country produces output of tradables and non-tradables with a constant-return production function

$$Y_T = A_T F(K_T, L_T) \quad (3)$$

$$Y_N = A_N F(K_N, L_N) \quad (4)$$

where K is capital, L is labor, and A is productivity. Since labor is immobile between countries but mobile across the two sectors within a country, the same wage rate w applies to both sectors. Total domestic labor supply is $L = L_T + L_N$, so that the two sectors compete for labor. There is no similar constraint for capital, as capital is internationally mobile. Moreover, a common assumption is that productivity differs only in the tradable sector, whereas there are no productivity differentials in the non-tradable sector across countries (e.g., a barber in New York is not more productive than a barber in London).

A standard result in this simple type of model is that the relative price of non-tradables to tradables, p (treating tradables as the numeraire), is given by

$$\hat{p} = \frac{\mu_{LN}}{\mu_{LT}} \hat{A}_T - \hat{A}_N \quad (5)$$

where hats denote percentage changes (log differentiation, $\hat{Z} = dZ/Z$), $\mu_{LN} = wL_N/pY_N$ and $\mu_{LT} = wL_T/Y_T$ denote the labor share of income generated in the non-tradable and tradable sectors, respectively, and $\hat{A}_i$ denotes productivity changes in sector i . A reasonable assumption is that $\mu_{LN}/\mu_{LT} \geq 1$ because non-tradables are typically more labor intensive than tradables.

Let the price level of a country be a simple geometric average with weights γ for tradables and $1 - \gamma$ for non-tradables. Each country produces a tradable of a certain quality with price P_T that is proportional to its quality X , $P_T = X$, and a non-tradable with price P_N . As

above, the relative price of non-tradables, P_N/P_T is denoted by p . Then the overall price levels at home and abroad ($\star$) are given by

$$P = X^\gamma P_N^{1-\gamma} = X p^{1-\gamma} \quad (6)$$

$$P^\star = (X^\star)^\gamma (P_N^\star)^{1-\gamma} = X^\star (p^\star)^{1-\gamma}. \quad (7)$$

The PPP rate $P/P^\star$ is then given by

$$\frac{P}{P^\star} = \left(\frac{X}{X^\star} \right) \left(\frac{p}{p^\star} \right)^{1-\gamma} \quad (8)$$

and log-differentiating and substituting the expression in Eq. (5) for $\hat{p}$ and $\hat{p}^\star$ yields

$$\hat{P} - \hat{P}^\star = \underbrace{(\hat{X} - \hat{X}^\star)}_{\text{Quality}} + (1 - \gamma) \underbrace{\left[\frac{\mu_{LN}}{\mu_{LT}} (\hat{A}_T - \hat{A}_T^\star) - (\hat{A}_N - \hat{A}_N^\star) \right]}_{\text{HBS}}. \quad (9)$$

Note that under the assumption that $\hat{A}_N - \hat{A}_N^\star = 0$ discussed above, the only sources of cross-country price level variation are productivity differentials in the tradable sector (HBS effect) and quality differentials in the tradable sector (Quality).

Given the definition of the RER $Q_t = \frac{P_t}{S_t P_t^\star}$, it is clear that a lower quality of tradable goods abroad will be generally accompanied by a weaker real exchange rate. Likewise, under the assumption that $\mu_{LN}/\mu_{LT} \geq 1$, i.e. non-tradables are more labor intensive than tradables, a higher relative productivity in the U.S. (tradables to non-tradables) compared to the foreign country will result in a higher RER, i.e. the foreign currency will be weaker in real terms.

Empirically, the HBS or Penn effect is often assessed by simple cross-sectional regressions of real exchange rates on the level of real *per capita* GDP across countries (at a given point in time). We also rely on this approach – which fits naturally into our cross-sectional focus on currency value portfolios – but augment the regression by a measure of export quality (since tradables in the above framework can be thought of as export goods), to obtain

$$q_{j,t} = \alpha_t + \beta_t GDP_{j,t} + \gamma_t Qual_{j,t} + \nu_{j,t}, \quad (10)$$

where $j = 1, \dots, N$ indexes countries and the residual ν_j is intended to measure the over- or

undervaluation of the currency after accounting for the Penn effect and product differentiation due to quality effects.⁵ The residuals in the above regression are then informative about (relative) exchange rate over- or undervaluation after correcting for these structural differences across countries.

2.3 Decompositions based on present-value relations

To shed further light on the information content in the RER and its economic drivers we also draw on present-value formulations of real exchange rates (e.g. Froot and Ramadorai, 2005; Engel and West, 2005).

We start with a decomposition similar to Froot and Ramadorai (2005). By definition, the currency excess return, rx , is

$$rx_{t+1} = -(q_{t+1} - q_t) + (i_t^* - i_t) - (\pi_{t+1}^* - \pi_{t+1}), \quad (11)$$

where π is the change in price (inflation), and i denotes the interest rate (Froot and Ramadorai, 2005). Rewrite as

$$q_t = rx_{t+1} - (i_t^* - i_t) + (\pi_{t+1}^* - \pi_{t+1}) + q_{t+1}, \quad (12)$$

and then take conditional expectations and iterate forward to obtain

$$q_t = \sum_{h=1}^{\infty} E_t [rx_{t+h}] - E_t [i_{t+h-1}^* - i_{t+h-1}] + E_t [\pi_{t+h}^* - \pi_{t+h}], \quad (13)$$

where we impose $\lim_{h \rightarrow \infty} q_{t+h} = 0$, i.e., the RER level is unity in expectation in the very long run. Hence, the RER can be decomposed into (i) *expected* future excess returns, i.e., currency risk premia (or, alternatively, deviations from uncovered interest parity), (ii) the *current* and future *expected* interest rate differential, and (iii) *expected* future inflation differentials.

Similarly, Engel and West (2006) set up a decomposition of the RER that incorporates a

⁵Frankel (2006) uses similar regressions (without the quality term) to measure Renminbi misalignments, consistent with a large literature on this topic (see Taylor and Taylor, 2004, and the references therein).

link to fundamentals underlying the monetary policy decision of central banks.⁶ Specifically, they posit that central banks in the home and foreign country follow a Taylor rule (Taylor, 1993), thereby linking the RER to expectations about future relative inflation rates, output gaps, and risk premia. A slightly modified version of their setup assumes that the central banks of the two countries follow an identical Taylor rule

$$i_t = \gamma_\pi E_t \pi_{t+1} + \gamma_y y_t \quad (14)$$

$$i_t^* = \gamma_\pi E_t \pi_{t+1}^* + \gamma_y y_t^* \quad (15)$$

where y denotes the output gap, and uncovered interest rate parity, in logs, holds up to a risk premium ρ (which is identical to the expected excess return $E_t(rx)$ above)

$$i_t - i_t^* = E_t s_{t+1} - s_t - \rho_t. \quad (16)$$

Subtracting Eq. (14) from (15), inserting into (16), adding expected inflation differentials $E_t \pi_{t+1}^* - E_t \pi_{t+1}$ on both sides of the equation and rearranging yields

$$q_t = (1 - \gamma_\pi) (E_t \pi_{t+1}^* - E_t \pi_{t+1}) - \gamma_y (y_t^* - y_t) + \rho_t + E_t q_{t+1}. \quad (17)$$

Iterating this equation forward, taking conditional expectations and imposing long-run PPP as above results in a present value formulation, suggesting that movements in real exchange rates are driven by current and future expected inflation, output gap differentials as well as the current and future expected risk premium.

We use the decompositions described above to measure the relevance of macroeconomic fundamentals (including interest rates, expected inflation, output gaps) in explaining the predictive power of the RER for the cross-section of currency (excess) returns.

⁶Taylor-rule fundamentals have been found to be accurate predictors of bilateral exchange rates in the macro exchange rate literature (e.g. Molodtsova, Nikolsko-Rzhevskyy, and Papell, 2008; Molodtsova and Papell, 2009).

3 Data

3.1 Data and currency returns

Exchange rates. Our exchange rate data are taken from the Global Financial Database (GFD) and cover a long sample period from 1970Q1 to 2014Q1 at a quarterly frequency. Exchange rates are end-of-quarter values. The empirical analysis is based on exchange rate changes and returns starting in the first quarter of 1976, i.e. shortly after the fall of Bretton Woods. We use the earlier years from 1970 to 1975 to compute long-run growth rates of various macro variables and RER changes to ensure that the empirical analysis can start in the first quarter of 1976.

Our RER measure is based on real exchange rates normalized to unity in 1970Q1 for all countries. We can proceed this way as we have a balanced sample of consumer price inflation indices (CPI) – also obtained from the GFD – and exchange rates spanning the entire sample period for all countries. However, these real exchange rates are based on relative PPP rates, which means that there is a base-year effect. Later on in the paper, we will show that our results are robust to this issue, by drawing on RERs computed from absolute PPPs.⁷

We collect data on a total of 23 developed and emerging countries. Since all exchange rates are quoted against USD, we have a total of 22 exchange rates. Our sample covers Australia, Canada, France, Germany, Hungary, India, Indonesia, Italy, Japan, South Korea, Mexico, New Zealand, Norway, Singapore, South Africa, Spain, Sweden, Switzerland, Taiwan, Thailand, Turkey, United Kingdom as well as the United States. We eliminate France, Italy, and Spain from the sample in 1999 after they adopted the Euro, only keeping Germany in the sample.

Macro fundamentals. In addition to spot rate data and CPI indices we also obtain data on additional (macro) variables for the same set of countries, retrieved from the GFD. These

⁷These data, obtained from the OECD, are available for a slightly smaller cross-section of countries and PPP rates are only available at the annual frequency but the results reported in the following are qualitatively robust to using this absolute PPP measure. We choose to work with GFD-based PPP and RER in our main analyses because of the larger cross-section, higher data frequency, and because the OECD PPP rates were constructed in the 1990s and were not available in the 1970s (the start of our sample).

data include short-term (3-month) interest rate, nominal GDP, and population figures. Not all macro variables are available for all countries and/or at each point in time, and hence we are dealing with an unbalanced panel. Where data from the GFD are not available we fill any missing data by drawing on other sources (primarily Thomson Datastream).

Currency returns. We are interested in the returns to value strategies and will generally consider two sets of returns in the empirical analysis that relies on portfolio returns. The first is the currency excess return which can be written as

$$RX_{t+1} = \frac{S_{t+1} (1 + i_t^*)}{S_t (1 + i_t)}$$

where S_t denotes the nominal exchange rate at the end of quarter t , i_t^* denotes the end-of-quarter foreign interest rate, and i_t denotes the end-of-quarter U.S. interest rate. The excess return measures the return accruing to a U.S. investor who borrows at the US interest rate i and uses the funds to hold a position in foreign currency for one quarter, earning the foreign interest rate i^* and then converting the proceeds back to dollars.

We also compare the results for currency excess returns to those of simple currency returns, or exchange rate changes, which just represent the spot rate component of the currency excess return (i.e., $\Delta S_{t+1} = S_{t+1}/S_t - 1$). While excess returns tell us about the overall return of a U.S. investor trading in foreign currencies, the exchange rate change isolates the return contribution from changes in the asset price itself, i.e., the change in the spot exchange rate. This distinction is important for the empirical analyses later in the paper.

Note that our definition of nominal spot exchange rates (USD/FC) means that a positive ΔS_{t+1} implies an appreciation of the foreign currency. Likewise, a positive RX means a positive excess return to holding foreign currency for a U.S. investor.

3.2 Descriptive statistics

Table 1 reports annualized mean excess returns and spot exchange rate changes as well as annualized volatilities for all exchange rates in our sample. We also report average RER levels and annualized quarterly mean RER changes as well as (annualized) means and volatilities

for CPI inflation rate differentials and nominal GDP growth rates.

As can be seen, there are some countries with positive appreciation rates (against the USD) on average; these countries include Japan, Germany, Singapore and Switzerland which are typically regarded as countries with fairly stable macroeconomic conditions. The same countries also feature low average excess returns, strong real exchange rates, and low inflation. We will return to this pattern below in our empirical analysis.

– Table 1 about here –

3.3 How do we measure the quality of export goods?

As outlined above, our fair value RER approach relies on two adjustments, i.e. HBS effects and differences in the quality of export goods. Whereas the first adjustment is well established in the literature (see Section 2.2), the second adjustment deserves further discussion. Regarding the export quality index we rely on data constructed by the International Monetary Fund and described in detail by Henn, Papageorgiou, and Spatafora (2013). The basis for the index construction is an extension of the UN-NBER dataset covering bilateral trade at the SITC 4-digit level. The quality index of export goods is constructed bottom up from a very disaggregated level of 851 product categories. The approach is to adjust unit values for differences in production costs and a selection bias coming from relative distance between exporter and importer. At the end of this process there are about 20 million quality estimates, each covering a specific exporter-importer-product combination. These estimates are then aggregated in a final step into a country index of export quality.

For our analysis below, we construct (log) export quality differentials (against the U.S.) and set the relative quality to a value of zero in 1970 in the same way as for our CPI-based real exchange rates, so that the two measures are comparable.

4 Predictive content of currency value

We rely on currency portfolios and cross-sectional regressions to gauge the predictive content of currency value for future currency (excess) returns. The portfolio design closely follows

the implementation of quantitative investment strategies by professional currency managers (see e.g., Pojarliev and Levich, 2012, for details about investment strategies pursued in this industry). A key advantage is that such a framework allows for a straightforward assessment of the economic value of predictability of exchange rate changes and excess returns.

4.1 Constructing currency value portfolios

In our benchmark setup, we build currency portfolios based on linear weights given by

$$w_{j,t+1} = c_t (x_{j,t} - \bar{x}_t), \quad (18)$$

where $x_{j,t}$ denotes the signal for currency j in quarter t (such as the RER) and $\bar{x}_t = N_t^{-1} \sum_{j=1}^{N_t} x_{j,t}$ denotes the *cross-sectional average* of this signal (across countries, N_t). c_t is a scaling factor such that the absolute sum of all portfolio weights equals unity. Currencies with a value of the signal above the cross-sectional mean receive positive portfolio weights, whereas currencies with a below-average value receive negative portfolio weights. The portfolio return rx^p is then given by $rx_{t+1}^p = \sum_{j=1}^{N_t} w_{j,t+1} rx_{j,t+1}$. In the implementation of this approach we re-balance the portfolios at the end of each quarter.

This setup where weights are linear in the signal is simple but very useful for decomposing the overall portfolio return into different components of the signal. For example, suppose we can decompose a signal $x_{j,t}$ into two components such that $x_{j,t} = x_{1,j,t} + x_{2,j,t}$; then the returns to the two portfolios based on $w_{1,j,t+1} = c_t(x_{1,j,t} - \bar{x}_{1,t})$ and $w_{2,j,t+1} = c_t(x_{2,j,t} - \bar{x}_{2,t})$ will add up to the overall portfolio return based on the composite signal $x_{j,t}$ defined above. This allows us to perform simple decompositions of the predictive information in currency value into different drivers (e.g., changes in spot rates vs inflation).

In addition to the benchmark results, we also report returns of rank portfolios (see, e.g. Asness, Moskowitz, and Pedersen, 2012; Koijen, Moskowitz, Pedersen, and Vrugt, 2013), whose weights are given by

$$w_{j,t+1} = c_t \left(\text{rank}(x_{j,t}) - \sum_{j=1}^{N_t} \text{rank}(x_{j,t}) / N_t \right). \quad (19)$$

The scaling factor c_t is analogous to the case of linear portfolio weights above and ensures that portfolio weights sum to one in absolute value. This procedure is more conservative (in that outliers and other extreme scores of signals receive a smaller weight), but the downside is that it does not permit exact decompositions.

Finally, we also perform standard cross-sectional portfolio sorts for comparison, sorting currencies into four bins (P_1, P_2, P_3, P_4) based on quartiles of the cross-sectional distribution of real exchange rates. Within each bin, currencies are equally weighted (as, e.g., in Lustig, Roussanov, and Verdelhan, 2011; Menkhoff, Sarno, Schmeling, and Schrimpf, 2012a). We report results for a high-minus-low portfolio (HML) which is long in P_4 (weak real exchange rates) and short in P_1 (strong real exchange rates).

Benchmark results. We report benchmark results for returns to currency value portfolios in Table 2. Currencies with relatively low valuation levels, i.e. weak RERs (high Q), receive positive weights in the portfolio, whereas currencies with relatively high valuation levels, i.e. strong RERs (low Q) receive negative weights (see Eq. (17)). In effect, we go long relatively “cheap currencies” and short relatively “expensive currencies”.⁸ Note that this cross-sectional approach does not rely on the absolute valuation level against the USD. For example, even if PPP indicates that all currencies are undervalued against USD, we would still go long the most undervalued currencies and short the least undervalued currencies.

The table shows that RERs significantly predict subsequent currency excess returns in the expected way, i.e. “cheap” currencies outperform “expensive” currencies. This result holds independently of the exact way currency value portfolios are built (linear weights, rank, equal weighting within bins). A portfolio taking long positions in currencies with weak RERs (high Q) funded by shorting strong RER currencies (low Q) is profitable with Sharpe Ratios between 0.27 and 0.35 per annum.

– Table 2 about here –

Interestingly, however, real exchange rates forecast future changes in nominal spot rates

⁸Using RER levels for forming portfolios is also done in the financial industry, e.g. in the RER-based Currency Valuation ETF of Deutsche Bank.

with a *negative* sign. In other words, low real currency valuations (high Q) tend to be followed by nominal depreciations. This predictability is statistically significant and sizeable in economic terms with average annualized exchange rate changes of -3.10 to -7.17%.

We will investigate this finding more closely below but note here that these results are not in line with the standard interpretation that RERs proxy for currency over-/undervaluation. Under such a story one would expect to see that the predictability comes (at least partly) from an adjustment of spot exchange rates. However, we find the opposite: Average excess returns of currency value portfolios are positive but this is clearly driven solely by the carry component of excess returns, not from spot exchange rates adjusting to PPP.

In the following, we explore the link between currency value and currency returns from various angles to provide a better understanding of the drivers of value and the determinants of currency risk premia in the cross-section of currencies.

4.2 Levels versus changes

We proceed with a simple decomposition of RERs into lagged levels and subsequent changes as outlined in Eq. (2) above

$$q_t = \underbrace{(p_{t-5y} - p_{t-5y}^* - s_{t-5y})}_{\text{Past level } q_{t-5y}} + \underbrace{(\pi_{t-5y;t} - \pi_{t-5y;t}^* - \Delta s_{t-5y;t})}_{\text{Change until today } \Delta q_{t-5y;t}}.$$

Here again a high q is associated with a weak RER to make sure that the decomposition directly translates into weights for currency value portfolios.

Motivated by this mechanical decomposition, we form portfolios separately based on the lagged level of the RER 5 years ago (q_{t-5y}) as well as the change in the RER over the past 20 quarters ($\Delta q_{t-5y;t}$). Here and in the remainder of the paper, we only look at portfolios based on linear weights so that we get exact decompositions.⁹ Using the change in the RER over the last five years is admittedly somewhat arbitrary but it roughly corresponds to business cycle frequency and, more importantly, the five-year change in the RER is also a common proxy for currency value in the literature (Asness, Moskowitz, and Pedersen, 2012; Kroencke,

⁹Our results remain qualitatively the same when using rank portfolios instead of linear portfolio weights.

Schindler, and Schrimpf, 2014; Barroso and Santa-Clara, 2014). We can decompose this long-run change in the RER further into two components, the negative of lagged nominal depreciation rates $-\Delta s_{t-5y;t}$, and the negative of the differential between foreign inflation and home (in our case U.S.) inflation $-(\pi_{t-5y;t}^* - \pi_{t-5y;t})$.

– Table 3 about here –

Table 3 shows that this simple decomposition uncovers interesting predictability patterns. The aggregate average excess return of 1.62% for the linear portfolio is made up of a 0.43% excess return due to the signal in lagged RER levels, and a 1.19% excess return due to the signal in five-year RER changes. Hence, almost 75% of the overall performance of currency value is attributable to five-year RER changes and less than 25% stems from lagged levels.

When looking at the predictive content for exchange rate changes, this pattern is even more striking. The overall exchange rate return of the linear-weights RER portfolio of -3.14% can be broken down into a -3.68% contribution from the information in five-year lagged RER levels (which thus accounts for more than 100% of the overall return), and a 0.54% contribution from the information in the five-year change in RERs (which therefore works against the overall predictability pattern). Hence, all the exchange rate predictability stems from information in RER levels and none comes from RER changes.

This means that RER levels and RER changes forecast excess returns for different reasons. RER levels must be strongly related to interest rate differentials (carry). Countries with weak real exchange rates must offer high carry to compensate investors for the nominal depreciation of the exchange rate. In contrast, five year RER changes forecast exchange rate changes and excess returns in line with the intuition of currency value: Currencies with strong long-term RER appreciations tend to depreciate, whereas currencies with long-term RER depreciations tend to appreciate - in line with the idea of a reversal effect. However, this link seems fairly weak as RER changes do not forecast exchange rate changes significantly. Moreover, a substantial share of the overall excess return does not come from adjusting exchange rates (1.19% overall excess return compared to only 0.54% from exchange rate changes). Hence, even for five-year RER changes, most of the excess return predictability comes from the carry component, while the adjustment of spot rates only drives less than half of the overall excess

return predictability.

We can further split up the predictive content of *five-year changes in real exchange rates* according to the contribution from (the negative of) five-year spot exchange rate changes and (the negative of) inflation differentials between the foreign country and the U.S. The average excess return of the portfolio constructed from five-year RER changes comes entirely from the signal derived from (the negative of) 5-year spot exchange rate changes (which has an average return of 5.87%). Currencies with strong past depreciations (i.e. a large $-\Delta s$) tend to yield high future excess returns, i.e. command a high risk premium. Lagged CPI inflation differentials (i.e. the negative inflation differential), by contrast, significantly predict excess returns with a negative sign (-4.68%): High inflation countries command higher risk premia on average than low inflation countries. Thus, the positive predictive ability of RER changes – as in, e.g., Asness, Moskowitz, and Pedersen (2012) – for excess returns stems exclusively from the predictive content in long-run spot rate changes. The inflation component of RER changes attenuates predictability.

How can we interpret the predictability of excess returns by lagged spot rate changes? It seems tempting to regard this phenomenon as a reversal effect where strong appreciations are followed by depreciations going forward. This would be akin to similar kinds of reversal effects in equity markets, as demonstrated by DeBondt and Thaler (1985), and the findings in the extant literature on currency value strategies have generally been interpreted like this. However, results for exchange rate changes in Table 3 indicate that lagged spot rate depreciations ($-\Delta s_{t+1}$) actually predict future spot exchange rates with a negative sign, i.e., depreciations over the past five years are followed by further depreciations on average.

This finding, taken together with the result that high relative valuation levels forecast exchange rate appreciations, seems inconsistent with a story of mean-reverting asset prices and currencies adjusting in response to over-/undervaluation. Instead, a more plausible explanation is that currencies of a particular set of countries persistently appreciate, whereas those of other countries persistently depreciate *relative* to each other in the cross-section of countries. Countries with expensive currencies (low Q) exhibit a low risk premium, whereas countries with cheap currencies (high Q) feature a high risk premium. This finding is consistent with our initial observation from Table 1 above that countries such as Japan, Germany,

Singapore and Switzerland tend to have appreciating exchange rates on average (in nominal terms), low risk premia, and strong real exchange rates.

The remaining entries in Table 3 show that the negative exchange rate predictability by RER levels entirely stems from lagged RER levels. Lagged inflation differentials predict future exchange rate returns with a positive sign, i.e. high inflation countries feature depreciating exchange rates. The magnitude of the latter effect is remarkable in light of the difficulties in the classical exchange rate literature to detect a robust link between macro fundamentals and movements in spot exchange rates (e.g. Meese and Rogoff, 1983). This also shows the virtues of a multi-currency and cross-sectional perspective instead of purely looking at bilateral exchange rates through the lens of time-series regressions.

– Figure 1 about here –

– Figure 2 about here –

Figures 1 and 2 depict cumulative returns for excess returns (and the carry component) and spot exchange rate changes, respectively. These cumulative returns are fairly smooth, again indicating the strength of persistent differences across countries. We also report results for a carry portfolio for comparison.

Cross-sectional regressions. We next turn to cross-sectional regressions, in the spirit of Fama and MacBeth (1973), of currency (excess) returns on (lagged) measures of currency value. This serves as an additional exercise (beyond the portfolio sorts) to gauge the predictive content in valuation measures for excess returns in the cross-section of currencies.

Table 4 reports results for cross-sectional regressions of excess returns (left part of the table) and exchange rate changes (right part) on (lagged) RER levels, five-year RER changes as well as (the negative of) five-year spot exchange rate changes and inflation differentials. When including five-year RER changes and lagged five-year levels simultaneously, we find that five-year RER changes appear to be more successful in predicting excess returns, while lagged levels mostly matter for exchange rate changes. This finding is in line with our results above and again suggests that signals based on five-year RER changes largely eliminate the

persistent cross-country characteristics driving exchange rate predictability by real exchange rates. It also corroborates our finding above that portfolios based on RER changes are not fully driven by adjusting spot exchange rates.

– Table 4 about here –

When considering both lagged RER levels as well as (the negative of) lagged five-year exchange rate changes and inflation differentials simultaneously as regressors (in line with the decomposition in Eq. (2)), we find that only lagged spot rate changes predict future excess returns. Consistent with our previous results, long-term currency depreciations predict high risk premia in the cross-section of countries. Inflation differentials are insignificant and have the opposite sign compared to the portfolio results in Table 3. Inflation differentials, by contrast, capture all the predictability for future exchange rate changes. The link is such that higher inflation abroad over the previous five years predicts a relative depreciation of the currency going forward. Lagged 5-year spot exchange rate changes also enter significantly, albeit with a different sign compared to the portfolio results in Table 3.¹⁰

The economic magnitudes seem quite large, in line with the portfolio returns documented above. For example, an increase in the 5-year RER change of one corresponds to a quarterly increase in the expected excess return of 2.58%. Table 1 shows that the typical standard deviation of 5-year RER changes is about 30% (annualized), which corresponds to a 15% quarterly standard deviation of the 5-year RER change. Hence, the regression estimate implies a $0.15 \cdot 2.58 = 0.39\%$ increase in expected quarterly excess returns for a currency with a one standard deviation higher 5-year RER change relative to the other currencies. Similarly, the economic magnitudes of 5-year spot exchange rate changes and inflation differentials point towards a high degree of economic significance.

Overall, we find that the classical measure of currency value, the RER, contains significant information for future currency returns. Not only does the RER level capture variation in risk

¹⁰The “wrong” signs for 5-year inflation differentials in the excess return regressions (left panel of Table 4) and for 5-year spot exchange rate changes in the exchange rate regressions (right panel of Table 4) are driven by collinearity issues since 5-year relative inflation rates and 5-year exchange rate changes are highly correlated in the cross-section. If we run the same regressions as in Table 4 but include only one of the two variables at a time, we find that lagged 5-year exchange rate changes are significant drivers of future excess returns, whereas 5-year inflation rate changes are significant drivers of future exchange rate changes.

premium, it is also informative about future changes in nominal spot exchange rates, albeit not as in a standard currency value concept. Currencies with low valuations (high Q) feature high excess returns going forward. Risk premia required by investors for exposure to particular currencies thus vary with currency value in the cross-section.

Interestingly, low valuation levels also predict future nominal depreciation. Our evidence thus suggests that currency risk premia can be regarded as fairly persistent and mostly driven by strong asymmetries across countries. Longer-term trends of appreciation tend to be followed by even more appreciation and signal low risk premia, while periods of high inflation tend to be followed by depreciations and signal high risk premia.

5 Currency fair value and macroeconomic fundamentals

This section extends the former, more conventional examinations of predictability by using the above introduced fair value concept. Moreover, we complement the portfolio sorts by cross-sectional regressions which allow us to control for competing drivers of returns.

5.1 Currency fair value

We construct measures of currency value that adjust the RER for the persistent cross-country differences that appear to drive our results reported above. To implement this idea, we conceptually follow ideas from the literature on “currency fair value” (e.g. Cenedese and Stolper, 2012) and relate the RER to measures that capture persistent deviations from PPP across countries. Drawing on the simple theoretical motivation in Section 2.2, we rely on differentials in productivity and quality of export goods to capture these slow-moving levels of real exchange rates.

Specifically, for each quarter in our sample we estimate a cross-sectional regression of real exchange rates on a measure of Harrod-Balassa-Samuelson effects (HBS) and a quality measure of a countries’ exports ($Qual$), as described in Section 3. The HBS measure is simply given by real per capita GDP. The cross-sectional regression reads

$$q_{j,t} = a_t + b_{1,t}HBS_{j,t} + b_{2,t}Qual_{j,t} + u_{j,t}. \quad (20)$$

We then form value portfolios based on the signal in q_t and each of its components $\hat{b}_{1,t}HBS_{j,t}$, $\hat{b}_{2,t}Qual_{j,t}$, and the residual $\hat{u}_{j,t}$. In this framework, the residual $\hat{u}$ is intended to capture over-/undervaluation relative to intrinsic value (see, e.g., Frankel, 2006; Cheung, Chinn, and Fujii, 2007, for a similar approach). A high $\hat{u}_j$ indicates an undervaluation of the currency relative to fundamentals.¹¹

– Table 5 about here –

Results for this decomposition are shown in Table 5. The upper panel reports average excess returns and exchange rate changes for portfolios based on linear weights. Similar to the results in Table 2, we find an excess return for the portfolio based on RER levels of 1.71% p.a.¹² Roughly one half of this overall return can be attributed to the *HBS* and *Qual* components (mean returns of 0.16% and 0.72%, respectively), whereas the residual part (denoted *RES*) has a mean return of 0.82% and is highly significant. Both the HBS- and quality-related RER components significantly predict changes in nominal exchange rates with a negative sign. For both excess returns and exchange rate changes, the residual term, which captures over-/undervaluation effects net of these structural factors, significantly predicts returns with a *positive* sign.

The latter finding is important to understand our result above that unadjusted (low) RER levels predict a depreciation of nominal exchange rates (see e.g. Table 2). This exchange rate predictability by RER levels (lower valuation levels predict further depreciation) appears to be entirely driven by the HBS effect and differences in the quality of export goods across countries. Once we account for these cross-country differences, the remaining share of RER

¹¹The quality index is obtained from the IMF and is available at annual frequency. To convert it to quarterly frequency, we use the quality index in year t in all four quarters of the subsequent year. The data are available at <https://www.imf.org/external/np/res/dfidimf/diversification.htm>.

¹²The IMF quality data are available from 1970 until 2010. Since we use the value of the quality index in year t for all quarters of the following year $t+1$ (to go from annual to quarterly frequency in an out-of-sample fashion), our portfolio returns end in 2012Q1 (instead of 2014Q1 as in the full sample). In addition, there are no data for Taiwan, which we consequently drop from the sample for this exercise. Hence, the overall excess return differs slightly from the one presented in Table 2.

levels predicts future exchange rates positively. Put differently, currency “undervaluation” as measured by the residual in the above regression $\hat{u}_j$, is followed by an appreciating exchange rate. This is in line with a reversal effect, i.e. currency value in the original sense. Moreover, the average exchange rate change of 0.94% in Panel A of Table 5 (more than) fully accounts for the overall excess return of this portfolio (0.82%). Hence, excess return predictability here comes almost entirely from adjusting real exchange rates. This finding is quite intuitive and very different from what we find for unadjusted measures of currency value based on RER (i.e., PPP) alone.

Moreover, we find a similar pattern for five-year RER changes. Contrary to our findings in Tables 3 and 4 where 5-year RER changes do not significantly forecast exchange rates, we find that the adjusted RER measure now significantly forecasts both excess returns and exchange rate changes. Similar to the results for adjusted RER levels, the average excess return (2.56%) is (more than) fully driven by changes in exchange rates (2.75), in line with the standard intuition of currency value strategies.

We finally illustrate the difference between HBS and quality components of RER levels on the one hand, and the residual component of RER levels on the other hand, in forecasting excess returns and simple exchange rate changes in Figure 3. This figure shows post-formation portfolio returns for 1, 2, ..., 24 quarters after portfolio formation (linear portfolio weights) for both excess returns (left panels) and exchange rate changes (right panels), separately for portfolios based on the HBS component of RER, the quality component, and the “fair value” component (i.e., the residual in Equation (20) above).

– Figure 3 about here –

As can be seen from these plots, both the HBS and quality component of RER show a fairly static forecasting power for excess returns and exchange rates in that they forecast a more or less permanently negative excess return and a positive exchange rate change (appreciation). The predictive content of the fair value component, by contrast, behaves quite differently. Figure 3 shows that “undervalued” countries tend to have significantly positive excess returns, which are fully driven by appreciating exchange rates for about 6 quarters. The predictive power levels out after about 12 quarters. This again suggests that the fair value component

is more closely aligned with the notion that RERs forecast returns because of an adjustment in spot exchange rates to fundamental levels.¹³

Overall, the evidence in this section suggests that there is an important role for adjusting currency value measures for HBS and quality effects, and this also applies for five-year RER changes, even though they are less prone to persistent country characteristics by construction.

5.2 Present value relations and macro fundamentals

Having established a meaningful link between real exchange rates and currency (excess) returns, we now go further by isolating the information content of underlying macroeconomic determinants of real exchange rates. We do so in the context of value portfolios and by means of cross-sectional regressions.

Empirical setup. The general setup is that we regress real exchange rates at time t on (proxies for) expectations about macro fundamentals in the *cross-section* of currencies. At each point in time, we then use the estimated slopes to split up real exchange rates into components due to the different macro fundamentals and study the relative contribution to value returns.

First, consider the framework by Froot and Ramadorai (2005) in Eq. (13) above which suggests that expectations about future interest rate differentials and inflation differentials are the macro drivers of the RER. Also, recall that the present-value relation based on Taylor rule fundamentals in Eq. (17), which suggests that expected output gaps and inflation differentials are key macro drivers of the RER. Motivated by these two present-value expressions, we run cross-sectional regressions of RER on expected interest rate differentials, expected inflation differentials and expected output differentials, i.e.

¹³We provide the same set of plots for portfolios and decompositions based on 5-year changes in RER in Figure A.1 in the Internet Appendix.

$$q_{j,t} = a_t + b_{1,t} \left(\sum_{h=1}^{\infty} \hat{E}_t [\Delta^* i_{t+h-1}] \right) + b_{2,t} \left(\sum_{h=1}^{\infty} \hat{E}_t [\Delta^* \pi_{t+h}] \right) + b_{3,t} \left(\sum_{h=1}^{\infty} \hat{E}_t [\Delta^* y_{t+h}] \right) + u_{j,t}, \quad (21)$$

with country differentials denoted by Δ^* , so that $\Delta^* i = i^* - i$, $\Delta^* = \pi^* - \pi$ and $\Delta^* = y^* - y$.¹⁴

To run these cross-sectional regressions, proxies for expectations about future fundamentals are required. These could be obtained in various ways, e.g., by drawing on surveys of market participants or backing out implied expectations from vector autoregressions. For our benchmark results, however, we choose to work with an approach that is easy to implement at a given point in time for a broad set of countries. We approximate expectations as discounted long-run growth rates of some macro fundamental

$$\tilde{g}_t = \frac{\sum_{j=0}^{\infty} \phi^j g_{t-j}}{\sum_{j=0}^{\infty} \phi^j}, \quad (22)$$

and then use $\tilde{g}_t$ as a proxy for investors' long-run expectations about the fundamental g . For instance, $\tilde{\pi}$ will be our proxy for long-run inflation expectation based on the discounted sum of lagged CPI inflation rates. This approach to proxying for expectations has been used in earlier work on U.S. inflation expectations (e.g., Piazzesi and Schneider, 2011; Cieslak and Povala, 2015).¹⁵ For our setup based on quarterly data, we follow Piazzesi and Schneider (2011) and set $\phi = 0.98$. We truncate the sum at 20 quarters.

Moreover, for the decomposition we need relative output gap differentials (against the U.S.). We estimate output gaps by means of quadratic trend regressions (see, e.g., Clarida, Gali, and Gertler, 2000; Cooper and Priestley, 2009). We run these regressions separately for each country and on a recursively expanding window, allowing for an initialization window from 1970Q1 to 1975Q4.

¹⁴The present-value decompositions discussed above suggest to also include expectations of future excess returns. We omit this from the equation as it is not possible to measure expected excess returns for individual currencies in a reliable way.

¹⁵This method draws on insights from the adaptive learning literature (Evans and Honkapohja, 2009). In adaptive learning, agents learn recursively as soon as new data becomes available; $1 - \phi$ is the gain parameter. Kozicki and Tinsley (2005) find that long-run weighted averages for U.S. inflation match survey inflation expectations for the U.S. quite well.

Results for portfolios. Table 6 reports results for currency portfolios constructed this way. For the sake of comparability, the first column reports the average excess return (exchange rate change) for the benchmark portfolio based on real exchange rate levels.¹⁶ The next columns report returns attributable to the predictive signal in the three drivers of real exchange rates.

– Table 6 about here –

We find that the part of RER levels attributable to future output gaps contributes significantly to excess return predictability. In other words, the only macro fundamental that is significantly linked to excess return predictability by the RER is the (expected) output gap differential. For exchange rate changes, we find that both expected interest rate differentials and expected inflation differentials are significant – both statistically and in economic terms. Moreover, the finding that carry forecasts spot exchange rate changes with the opposite sign compared to excess returns is also well in line with findings in the earlier literature (e.g., Lustig, Roussanov, and Verdelhan, 2011; Menkhoff, Sarno, Schmeling, and Schrimpf, 2012a). Similarly, expected inflation differentials forecast spot rate changes with a positive sign, which is the opposite of what we find for excess return predictability.

Overall, excess return predictability by the RER seems to be driven by a business cycle fundamental – output gaps – which seems to make sense from the viewpoint of standard asset pricing models and previous findings in the literature (e.g. Lustig, Roussanov, and Verdelhan, 2014). Exchange rate predictability by the RER, however, is related to standard exchange rate fundamentals such as the interest rate and inflation (Engel and West, 2005; Sarno and Schmeling, 2014).

Regression results. We complement the above portfolio perspective and provide results for the underlying regression directly. Table 7 presents cross-sectional regressions of real exchange rates on macro fundamentals from the present value-decomposition and we also add HBS and quality as regressors for completeness.

¹⁶The average returns here differ somewhat from those reported in Table 3 because we do not have available macro fundamentals for all countries from the beginning of the sample. Hence, contrary to the benchmark results in Table 2 we are dealing with an (unbalanced) panel with fewer observations here so that portfolio returns are slightly different.

The first six specifications always include one macro component separately, specification (vii) is motivated by the decomposition in the spirit of Froot and Ramadorai, specification (viii) draws on the Taylor rule decomposition, and specification (ix) corresponds to the fair value approach.¹⁷

– Table 7 about here –

We find a significant role for all fundamentals in the univariate regressions with R^2 s ranging from 15% to 46%. Hence, there is a strong link between RER levels and underlying macroeconomic fundamentals. As expected, there is a positive correlation between interest differentials and real exchange rates, i.e. countries with weak real exchange rates feature high interest rate differentials (e.g. Heyerdahl-Larsen, 2014). In the joint regressions (specifications (vii) – (ix)) we find that current interest rate differentials, expected output gaps, HBS effects and quality matter most for cross-sectional variation in currency value.

Taken together, we find a significant and economically meaningful link between currency value and fundamentals in the cross-section of currencies. These findings, coupled with the predictive power of RER measures for exchange rates and excess returns, suggest that there is a significant role for macroeconomic fundamentals in exchange rate determination in the cross-section of currencies.

6 Additional results and robustness

6.1 Absolute PPP rates

The RER levels in our baseline analysis are based on normalizing real exchange rates to unity in 1970Q1. While normalizing RER levels in the early 1970s is common in the literature, the exact date is admittedly arbitrary. For robustness, we also compute portfolio returns based on an RER measure immune from base year problems. These data (taken from the OECD) are based on absolute PPP rates and hence are not plagued by a base year problem. The

¹⁷We split up the expected interest rate differential term in the Froot/Ramadorai decomposition into the current interest rate differential (Δ^*i) and expected future interest rate differential ($E[\Delta^*i]$) here since we want to test for the contemporaneous link between carry and real exchange rate levels directly.

downside is that the OECD data are available at annual frequency only and cover a smaller set of currencies. Due to data limitations of the OECD data, we need to rebalance currency value portfolios at an annual rebalancing frequency.

– Table 8 about here –

Table 8 reproduces Table 2, but relies on the alternative value measure based on the OECD data. We find very similar results in terms of average excess returns, exchange rate changes as well as Sharpe Ratios.¹⁸ Exchange rate predictability by RER levels is actually even stronger, when judged from the magnitude of Sharpe Ratios. Thus, this key earlier result is not driven by base year effects. We still find that both RER levels and five-year RER changes predict excess returns going forward. As before, we find that strong real exchange rates predict nominal appreciations. Long-term changes in real exchange rates, by contrast, predict excess returns. Overall, our main results are qualitatively unaffected when considering this alternative RER measure free of base year effects.

6.2 Carry, momentum, and (fair) value

We next examine whether a value portfolio based on HBS and quality effects contains useful information over and above that contained in standard currency value measures as well as currency carry and currency momentum.¹⁹

Table 9, Panel A, shows exposure regressions where we regress the excess returns of different value portfolios on a carry factor and a momentum factor. The upper three rows of Panel A report unadjusted mean returns and Sharpe Ratios for comparison while the lower part shows the exposure regression results and the information ratio (IR). We find that the portfolios based on 5-year RER changes ($\Delta q_{t-5y;t}$) and the value portfolio adjusted for HBS and quality effects ($\Delta q_{t-5y;t}^{RES}$) have significant alphas even after accounting for carry and momentum. Moreover, these portfolios have a negative carry beta. Hence, a value

¹⁸The only noteworthy difference in results occurs for the HML portfolio of the cross-sectional sort whose average excess return is no longer statistically significant in the case of RER levels. However, both average excess returns of the portfolio based on linear weights and the one based on the rank method remain significant.

¹⁹We rely on a currency momentum factor with a three-month formation and a one-month holding period (see Menkhoff, Sarno, Schmeling, and Schrimpf, 2012b).

strategy that goes long in undervalued currencies (weak real exchange rates), funded by short positions in overvalued currencies (strong real exchange rates), generates significantly positive alphas and a negative correlation with carry returns, which seems highly attractive from the viewpoint of investors (also see Asness, Moskowitz, and Pedersen, 2012, for results on the link between carry and value in currency returns).

– Table 9 about here –

We then run a regression of the HBS and quality adjusted value portfolio's (which we call "fair value" below for ease of exposition) excess returns on carry, momentum, and the "standard value" portfolio (based on 5-year changes in RER *without* adjusting for HBS and quality) excess return and obtain the following result

$$rx_{t+1}^{\text{fair value}} = \underset{[2.30]}{1.06} - \underset{[-1.03]}{0.03} rx_{t+1}^{\text{carry}} + \underset{[0.53]}{0.02} rx_{t+1}^{\text{mom}} + \underset{[9.98]}{0.62} rx_{t+1}^{\text{value}}, \quad R^2 = 0.64$$

with t -statistics in brackets and with an Information Ratio of 0.43. Hence, the fair value portfolio still has a significant alpha of about 1% p.a. and a high Information Ratio after accounting for simple value measures.

Finally, Panel B of Table 9 shows weights for the tangency portfolio in a simple mean-variance framework with carry, momentum, value, and fair value strategies as the investment universe. All four strategies are set up like they would be traded in real world applications, except for the fact that we do not take into account transaction costs.²⁰ As can be seen, carry and fair value receive the highest overall weight (32% and 35%, respectively), followed by momentum (26%), and the simple currency value portfolio (7%).

6.3 Conditional and unconditional portfolios

We also compare our currency value portfolios (which are *conditional* in nature as we update portfolios quarterly based on the most recent value score) to *unconditional* value portfolios

²⁰Transaction costs are highly unlikely to have meaningful consequences for the returns from the currency value strategies studied here for several reasons. First, the generally low transaction costs that characterize the currency market do not impact much on returns from currency strategies when the rebalancing period is infrequent (in our case quarterly). Second, consistent with what one would expect from a currency value model, the evidence in the paper shows that positions are quite persistent, so turnover is low.

as in Lustig, Roussanov, and Verdelhan (2011). Unconditional portfolios are based on either (i) the full sample average of a value measure or (ii) the average value measure in the first part of the sample (1976Q1 – 1995Q1). The former case is not investable as it uses full sample information, whereas in the latter case we use the first half of the sample to compute portfolio weights that we apply only in the second half of the sample.

We report results in Table 10, where Panel A shows results for full sample averages and Panel B shows results for the case where we split the sample in half. We provide results for portfolios based on RER levels, 5-year RER changes, and the fair value measure which adjusts for HBS and quality effects. The results for excess returns show that using more recent information (conditional portfolio) is clearly beneficial for portfolios based on 5-year RER changes and the fair value measure. For RER levels, the results are more mixed. For the full sample average in Panel A, we also find that the conditional portfolio outperforms the unconditional portfolio, whereas there is basically no difference in Panel B. Since the results in Panel B refer to an investable strategy, it seems reasonable to conclude that risk premia captured by RER levels are fairly static in nature but that looking at 5-year RER changes or the fair value measure removes this static component.

– Table 10 about here –

Similarly, for exchange rate changes, we find that all three unconditional portfolios yield negative average exchange rate changes, both in Panel A and B, whereas only RER levels yield negative average exchange rate changes in the conditional portfolios.

6.4 Implementation lags

Many macro variables are not available to investors and policy makers in real time. Given that most of our results stem from persistent differences in macro characteristics across countries, however, it seems reasonable to expect that lagging macro variables by a few more quarters should not change our basic conclusions.

To check this intuition, we allow for implementation lags of one quarter (which is the benchmark case in all tables and figures above) up to 20 quarters to allow for the fact that

macro fundamentals are reported with lags and/or revised over time. We present average excess returns (with 95% confidence intervals) in Figure 4 and average exchange rate changes in Figure 5 for portfolios based on RER levels q_t , lagged RER levels q_{t-5y} , five-year RER changes $q_{t-5y;t}$, five-year spot exchange rate changes $\Delta s_{t-5y;t}$, five-year inflation differentials $\Delta^* \pi_{t-5y;t}$ as well as interest rate differentials (carry, for comparison).

– Figures 4 and 5 about here –

As expected, allowing for these implementation lags does not alter any of the conclusions drawn above. All signals that generate significant currency (excess) returns with a one quarter implementation lag also do so when allowing for lags of six quarters or more.

6.5 Return correlations

To better understand the relation between portfolio returns of different RER components, we report return correlations for different portfolios in Table A.I. We also include a carry portfolio for comparison purposes. Panel A reports correlations of excess returns, whereas Panel B reports correlations of spot exchange rate changes. We find that portfolio returns based on RER levels exhibit a high correlation with carry portfolios, consistent with the findings in Table 7.

The decomposition of currency value into its ultimate macro drivers is thus informative about why there are strong asymmetries across countries in terms of risk premia. That said, the return correlation of the portfolios based on carry and five-year changes in real exchange rates is much weaker. This is consistent with Asness, Moskowitz, and Pedersen (2012) and Kroencke, Schindler, and Schrimpf (2014) who find that portfolios based on currency carry and their measures of currency value are only mildly correlated.

When looking at portfolios dissecting the value signal into its underlying macro drivers, we find that return portfolios based on the RER are highly correlated (in absolute terms) with those based on lagged five-year spot exchange rate changes and inflation differentials. Moreover, portfolio returns based on the (negative of) lagged five-year spot exchange rate and portfolio returns based on (the negative of) lagged five-year inflation differentials are also

highly negatively correlated. This is due to the fact that exchange rate changes and inflation differentials are highly negatively correlated in the cross section.

6.6 Influential currencies

We also check whether a particular currency drives a large share of our value portfolio returns. To this end, we run a cross-validation exercise, where we eliminate one country from the sample at a time. Results for portfolios based on RER levels and 5-year changes in RERs are shown in Table A.II in the Internet Appendix. Portfolio excess returns and simple exchange rate changes are fairly stable in this exercise so that our results do not seem to be driven by any particular currency.

7 Conclusion

The valuation of currencies is of key importance to policy makers and investors alike, but empirical evidence on the properties and determinants of “currency value” is still scattered and largely incomplete. Nevertheless, currency value measures based on the real exchange rate are commonly used for important practical purposes, e.g. for the design of currency investment products and strategies.

We contribute to this literature by investigating the predictive content of real exchange rate levels, long-term changes in real exchange rates, and subcomponents of real exchange rates related to macroeconomic fundamentals. Our ultimate goal is to provide a better understanding of the link between currency valuation and risk premia in the cross-section of currencies. We do this in a unified, multi-currency setup that provides a straightforward assessment of the economic value of predictability from measures of currency value.

We show that both levels and long-term changes of real exchange rates have meaningful predictive power for currencies, but differ sharply in how they do so. Real exchange rate levels mainly predict nominal exchange rate changes, whereas real exchange rate changes predict future currency excess returns. The predictive information is quite persistent, indicating that static country characteristics play an important role as risk premium drivers. When adjust-

ing real exchange rates for structural differences across countries – Harrod-Balassa-Samuelson effect and export quality – we do indeed find that overvalued (undervalued) currencies depreciate (appreciate) significantly to revert towards fundamental value.

These findings suggest that it is not a reversal effect in nominal exchange rates that is behind the predictability from standard real exchange rate measures inspired by the purchasing power parity condition. Instead, real exchange rates need to be adjusted for structural country differences to uncover currency under- and overvaluation effects and a reversal to fundamental value. The corresponding investment strategy is thus more in line with currency value in its original sense.

We further show that cross-country differences in real exchange rate levels and changes can be ultimately traced to differences in macro fundamentals, in particular to differentials in interest rates and output gaps. These macro-related sub-components of real exchange rates also have significant predictive content for currency excess returns. We thus present encouraging evidence that standard macro fundamentals account for large parts of the asymmetries across countries that are behind the cross-country variation in risk premia.

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Table 1. Descriptive statistics

This table reports descriptive statistics for excess returns (RX), exchange rate changes (ΔS), the log RER (q), (log) RER changes (Δq), inflation differentials ($\Delta\pi$), and output growth differentials (Δy). All quantities and growth rates are differentials (against the USD/U.S.). Except for the (log) real exchange rate q , all other variables are in percent and annualized. The sample period is 1976Q1 – 2014Q1 and the frequency is quarterly.

	RX			ΔS			q			Δq			$\Delta\pi$			Δy		
	μ	σ		μ	σ		μ	σ		μ	σ		μ	σ		μ	σ	
Australia	2.68	7.89		-0.20	7.69		-0.14	0.06		-10.04	32.47		1.02	1.52		1.95	1.43	
Canada	1.41	4.55		-0.02	4.50		0.07	0.04		1.25	21.91		-0.08	0.90		0.23	1.38	
France	1.87	8.49		-0.28	8.38		-0.24	0.06		-7.06	37.05		0.60	1.05		0.54	1.53	
Germany	1.55	8.32		2.28	8.26		-0.26	0.06		-2.34	36.64		-1.50	1.23		-1.58	1.69	
Hungary	6.45	9.56		-3.18	8.85		-0.20	0.09		-21.48	32.88		5.67	3.85		4.79	6.67	
India	2.39	4.76		-4.68	5.05		0.48	0.10		28.98	28.15		3.74	3.58		6.80	6.53	
Indonesia	4.38	12.41		-6.45	13.76		0.51	0.16		35.65	57.33		5.10	5.41		11.26	8.19	
Italy	2.64	8.36		-3.11	8.34		-0.14	0.05		-3.98	33.70		3.92	1.72		4.58	2.08	
Japan	1.23	9.22		3.64	9.09		-0.72	0.07		-24.26	34.80		-2.31	1.22		-2.94	2.48	
Korea	4.03	8.26		-1.28	8.07		0.06	0.05		0.76	30.69		2.01	1.87		6.90	2.47	
Mexico	4.38	13.30		-15.48	13.40		0.05	0.07		4.42	41.38		17.66	8.53		19.24	10.90	
New Z.	7.69	8.72		0.16	7.88		-0.27	0.07		-17.42	37.23		1.85	1.93		1.76	5.30	
Norway	2.00	7.74		0.38	7.52		-0.33	0.05		-9.49	30.30		0.35	1.58		1.96	1.91	
Singapore	-0.02	3.78		1.91	3.69		-0.19	0.04		-5.43	24.10		-1.50	1.52				
S. Africa	0.32	10.38		-5.43	10.09		0.24	0.08		19.31	38.32		5.41	1.85		6.74	2.67	
Spain	2.72	8.53		-3.05	8.30		-0.49	0.07		-22.48	45.46		4.06	1.95		4.50	1.71	
Sweden	1.43	7.98		-0.36	7.96		-0.04	0.06		10.18	37.26		0.39	1.57		0.60	3.22	
Switzerland	1.13	9.04		3.67	9.21		-0.66	0.06		-26.65	36.51		-1.93	1.28		-2.09	1.58	
Taiwan	0.38	3.86		0.73	3.89		-0.29	0.05		-2.43	23.97		-0.88	2.35		2.42	1.82	
Thailand	0.97	6.82		-0.71	6.69		0.10	0.07		9.16	28.78		0.30	1.90		3.64	6.02	
Turkey	14.93	12.56		-28.18	12.18		0.01	0.09		-5.33	43.88		30.72	9.03		31.92	16.77	
U.K.	2.22	7.49		0.04	7.35		-0.32	0.05		-12.82	28.82		1.26	1.44		1.19	1.28	

Table 2. Currency value portfolios

This table reports descriptive statistics for currency portfolios based on real exchange rates. We report results for cross-sectional portfolio sorts where we sort currencies into 4 bins based on the cross-section of signals (portfolios $P_1, \dots, P_4$ and a high minus low portfolio, HML, long P_4 and short P_1), as well as a rank portfolio (PF^R), and a simple portfolio based on linear portfolio weights (PF^L ; weights are linear in the cross-sectional deviation of signals from the cross-sectional mean). The upper part shows results for currency excess returns whereas the lower part shows results for log spot exchange rate changes. Real exchange rates are defined such that a higher real exchange rate indicates a weaker foreign currency. Hence, the portfolios go long currencies with a relatively low valuation and go short currencies with a relatively high valuation. Excess returns and exchange rates are defined such that positive numbers mean a positive return on holding the foreign currency. t -statistics in brackets are based on Newey and West (1987) standard errors. The sample period is 1976Q1 – 2014Q1 and the frequency is quarterly.

	P_1	P_2	P_3	P_4	HML	PF^R	PF^L
Real exchange rates (base 1970Q1)							
Excess returns							
mean	1.53	2.30	2.38	3.86	2.33	1.22	1.62
t	[0.97]	[1.46]	[1.58]	[2.69]	[1.69]	[2.01]	[2.38]
std	8.81	9.32	7.50	7.56	8.64	3.90	4.60
SR	0.17	0.25	0.32	0.51	0.27	0.31	0.35
Exchange rate changes							
mean	0.16	-1.09	-2.81	-7.01	-7.17	-3.10	-3.14
t	[0.12]	[-0.68]	[-2.07]	[-3.71]	[-3.74]	[-3.70]	[-3.33]
std	8.41	9.06	7.00	8.47	9.41	4.15	5.12

Table 3. Decomposing the predictive content of currency value

This table reports results for real exchange rate decompositions. The first column repeats results for the real exchange rate “level” (q_t). The second and third columns show results for lagged levels (5-years lagged, q_{t-5y}) and five-year changes in real exchange rates ($\Delta q_{t-5y,t}$) which add up to the real exchange rate level in the first column. The fourth and fifth column report results for (the negative of) 5-year nominal exchange rate changes ($-\Delta s_{t-5y,t}$), which can be interpreted as depreciation rates of foreign currencies against the USD, and (the negative of) 5-year inflation differentials ($-\Delta^* \pi_{t-5y,t}$), which can be interpreted as the inflation differential between the U.S. and the foreign country. Real exchange rates are defined such that a higher real exchange rate indicates a weaker foreign currency. Hence, the portfolios go long currencies with a relatively low valuation and go short currencies with a relatively high valuation. Excess returns and exchange rates are defined such that positive numbers mean a positive return on holding the foreign currency. The sample period is 1976Q1 – 2014Q1 and the frequency is quarterly.

	q_t	q_{t-5y}	$\Delta q_{t-5y,t}$	$-\Delta s_{t-5y,t}$	$-\Delta^* \pi_{t-5y,t}$
Excess returns					
mean	1.62	0.43	1.19	5.87	-4.68
t	[2.38]	[0.64]	[2.73]	[3.46]	[-2.81]
SR	0.35	0.10	0.48	0.66	-0.55
Exchange rate changes					
mean	-3.14	-3.68	0.54	-13.61	14.15
t	[-3.33]	[-5.17]	[0.91]	[-5.41]	[6.33]

Table 4. Cross-sectional regressions: Currency value and predictability

This table reports results from cross-sectional regressions of future returns (excess returns or simple currency returns) on lagged real exchange rates (q_t) and/or subcomponents of real exchange rates (like spot exchange rate changes and inflation differentials). Real exchange rates are defined such that a higher real exchange rate indicates a weaker foreign currency. Excess returns and exchange rates are defined such that positive numbers mean a positive return on holding the foreign currency. The fourth and fifth row, respectively, correspond to (the negative of) 5-year nominal exchange rate changes ($-\Delta s_{t-5y,t}$), which can be interpreted as depreciation rates of foreign currencies against the USD, and (the negative of) 5-year inflation differentials ($-\Delta^* \pi_{t-5y,t}$), which can be interpreted as the inflation differential between the U.S. and the foreign country. Excess returns and exchange rate changes are measured in percent. t -statistics in brackets are based on Newey and West (1987) standard errors. The sample period is 1976Q1 – 2014Q1 and the frequency is quarterly.

	Excess returns			Exchange rate changes		
q_t	1.00			-2.11		
	[2.01]			[-3.39]		
q_{t-5y}	0.51	-1.19		-3.17	-1.55	
	[0.81]	[-1.90]		[-4.84]	[-2.41]	
$\Delta q_{t-5y;t}$	2.58			1.25		
	[2.51]			[1.15]		
$\Delta s_{t-5y;t}$		1.93				-1.14
		[3.68]				[-2.04]
$\Delta \pi_{t-5y;t}$		-11.81				22.49
		[-0.98]				[2.01]
R^2	0.13	0.23	0.34	0.13	0.21	0.36

Table 5. Currency fair value: HBS and quality

This table reports results for portfolios based on real exchange rate levels (q_t) and a RER decomposition into a part due to Harrod-Balassa-Samuelson effects (HBS), a part driven by export quality ($Qual$), and the remaining part (a regression residual, denoted RES) based on Equation 20. The first set of results in each Panel A refers to real exchange rate levels, whereas the second set of results in each panel refers to 5-year real exchange rate changes. Real exchange rates are defined such that a higher real exchange rate indicates a weaker foreign currency. Hence, the portfolios go long currencies with a relatively low valuation (or, equivalently, low income/productivity (HBS) and low export quality) and go short currencies with a relatively high valuation (or, equivalently, high income/productivity (HBS) and high export quality). Excess returns and exchange rates are defined such that positive numbers mean a positive return on holding the foreign currency. t -statistics in brackets are based on Newey and West (1987) standard errors. The sample period is 1976Q1 to 2014Q1 on a quarterly frequency.

	Excess returns				Exchange rate changes			
Panel A. Real exchange rate levels								
	q_t	q_t^{HBS}	q_t^{Qual}	q_t^{RES}	q_t	q_t^{HBS}	q_t^{Qual}	q_t^{RES}
mean	1.71	0.16	0.72	0.82	-3.32	-1.88	-2.39	0.94
t	[2.12]	[0.49]	[1.89]	[2.05]	[-3.04]	[-3.21]	[-4.01]	[2.32]
SR	0.33	0.08	0.28	0.32				
Panel B. Real exchange rate changes								
	$\Delta q_{t-5y;t}$	$\Delta q_{t-5y;t}^{HBS}$	Δq_t^{Qual}	$\Delta q_{t-5y;t}^{RES}$	$\Delta q_{t-5y;t}$	$\Delta q_{t-5y;t}^{HBS}$	Δq_t^{Qual}	Δq_t^{RES}
mean	2.57	-0.36	0.37	2.56	1.96	-0.05	-0.73	2.75
t	[2.62]	[-0.67]	[1.28]	[2.91]	[1.59]	[-0.06]	[-1.45]	[3.69]
SR	0.51	-0.13	0.18	0.64				

Table 6. Decomposing the real exchange rate

This table reports results for portfolios constructed based on real exchange rate decompositions as in Froot and Ramadorai (2005) and Engel and West (2006). $E[\cdot]$ denotes long-run expectations of a variable, Δ^*i denotes interest rate differentials, $\Delta^*\pi$ denotes inflation differentials, and Δ^*y denotes differentials in output gaps. All differentials of macro variables are against the U.S. Real exchange rates are defined such that a higher real exchange rate indicates a weaker foreign currency. Excess returns and exchange rates are defined such that positive numbers mean a positive return on holding the foreign currency. t -statistics in brackets are based on Newey and West (1987) standard errors. The sample period is 1976Q1 – 2014Q1 and the frequency is quarterly.

	q_t	$E[\Delta^*i]$	$E[\Delta^*\pi]$	$E[\Delta^*y]$
Excess returns				
mean	2.03	2.71	-5.00	2.09
t	[2.51]	[0.95]	[-1.47]	[3.01]
SR	0.43	0.17	-0.29	0.76
Exchange rate changes				
mean	-2.96	-14.08	12.75	-0.98
t	[-3.21]	[-3.62]	[3.34]	[-1.22]

Table 7. Cross-sectional regressions: Real exchange rates and macro fundamentals

This table reports results for cross-sectional regression of real exchange rates on macro fundamentals, where Δ^*i denotes the current nominal interest rate differential (foreign country against the U.S.), $E[\Delta^*i]$ denotes the expected future interest rate differentials, $\Delta^*\pi$ denotes inflation differentials (foreign country minus U.S.), Δ^*y denotes output gap differentials (foreign country minus U.S.), HBS denotes a Harrod-Balassa-Samuelson effect measure (real output per capita), and $Qual$ is a measure of countries' export quality. Differentials of macro variables are always for the foreign country minus the U.S. Excess returns and exchange rate changes are measured in percent. t -statistics in brackets are based on Newey and West (1987) standard errors. R^2 is the adjusted R^2 . Real exchange rates are defined such that a higher real exchange rate indicates a weaker foreign currency. The sample period is 1976Q1 – 2014Q1 and the frequency is quarterly.

	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)
Δ^*i	6.83 [6.43]						11.36 [3.09]		
$E[\Delta^*i]$		5.69 [7.93]					-4.74 [-0.81]		
$E[\Delta^*\pi]$			10.22 [4.68]				10.67 [1.36]	1.73 [0.88]	
$E[\Delta^*y]$				1.31 [7.89]				1.32 [4.60]	
HBS					-0.12 [-5.88]				-0.08 [-3.28]
$Qual$						-2.74 [-11.26]			-2.91 [-9.56]
R^2	0.18	0.15	0.16	0.17	0.19	0.46	0.36	0.22	0.57

Table 8. Value portfolios based on absolute PPP rates

This table reports is similar to Table 2 but here we employ absolute PPP rates from the OECD to construct real exchange rates. For these PPP series, we do not need to normalize the real exchange rate in some base year. We only update portfolio weights annually as OECD PPP rates are only available on an annual frequency. t -statistics in brackets are based on Newey and West (1987) standard errors. The sample period is 1976 – 2014.

	P_1	P_2	P_3	P_4	HML	PF^R	PF^L
Excess returns							
mean	0.88	2.82	5.18	2.96	2.07	2.13	2.27
t	[0.50]	[1.96]	[2.88]	[1.96]	[2.96]	[3.38]	[3.35]
std	9.89	8.11	8.68	7.77	4.47	4.08	4.20
SR	0.09	0.35	0.60	0.38	0.46	0.52	0.54
Exchange rate changes							
mean	0.35	-0.35	-9.39	-3.07	-3.43	-4.10	-4.22
t	[0.22]	[-0.25]	[-4.91]	[-2.17]	[-4.10]	[-5.33]	[-5.19]
std	9.85	7.96	8.40	7.40	5.07	4.42	4.55

Table 9. Carry, momentum, and (fair) value

This table reports results for exposure regressions of RER-based portfolio returns on carry and momentum benchmark portfolio returns in Panel A. The carry portfolio is based on 3-month interest rate differentials and momentum (Mom) is based on one-year lagged exchange rate changes. IR denotes the annualized information ratio (alpha divided by residual standard deviation). q_t denotes a portfolio based on RER levels, $\Delta q_{t-5y,t}$ denotes a portfolio based on 5-year RER changes, $-\Delta s_{t-5y,t}$ denotes a portfolio based on (the negative of) 5-year spot exchange rate changes (foreign currency depreciation rates), $-\Delta^* \pi_{t-5y,t}$ denotes a portfolio based on (the negative of) 5-year inflation rate differentials (inflation differential between the U.S. and foreign countries), $\Delta q_{t-5y,t}^{HBS}$ denotes the portfolio based on the HBS component of 5-year RER changes, $\Delta q_{t-5y,t}^{Qual}$ the portfolio based on the quality component of 5-year RER changes, and $\Delta q_{t-5y,t}^{RES}$ a denotes a fair value portfolio based on 5-year RER changes adjusted for HBS and quality effects. Panel B reports tangency portfolio weights when combining a carry, momentum, standard value (5-year RER changes), and fair value (based on the residual of a regression of RER changes on HBS and quality) portfolios. Real exchange rates are defined such that a higher real exchange rate indicates a weaker foreign currency. Hence, the portfolios go long currencies with a relatively low valuation and go short currencies with a relatively high valuation. Excess returns are defined such that a positive numbers means a positive return to holding the foreign currency. t -statistics in brackets are based on Newey and West (1987) standard errors. The sample period is 1976Q1 – 2014Q1 and the frequency is quarterly.

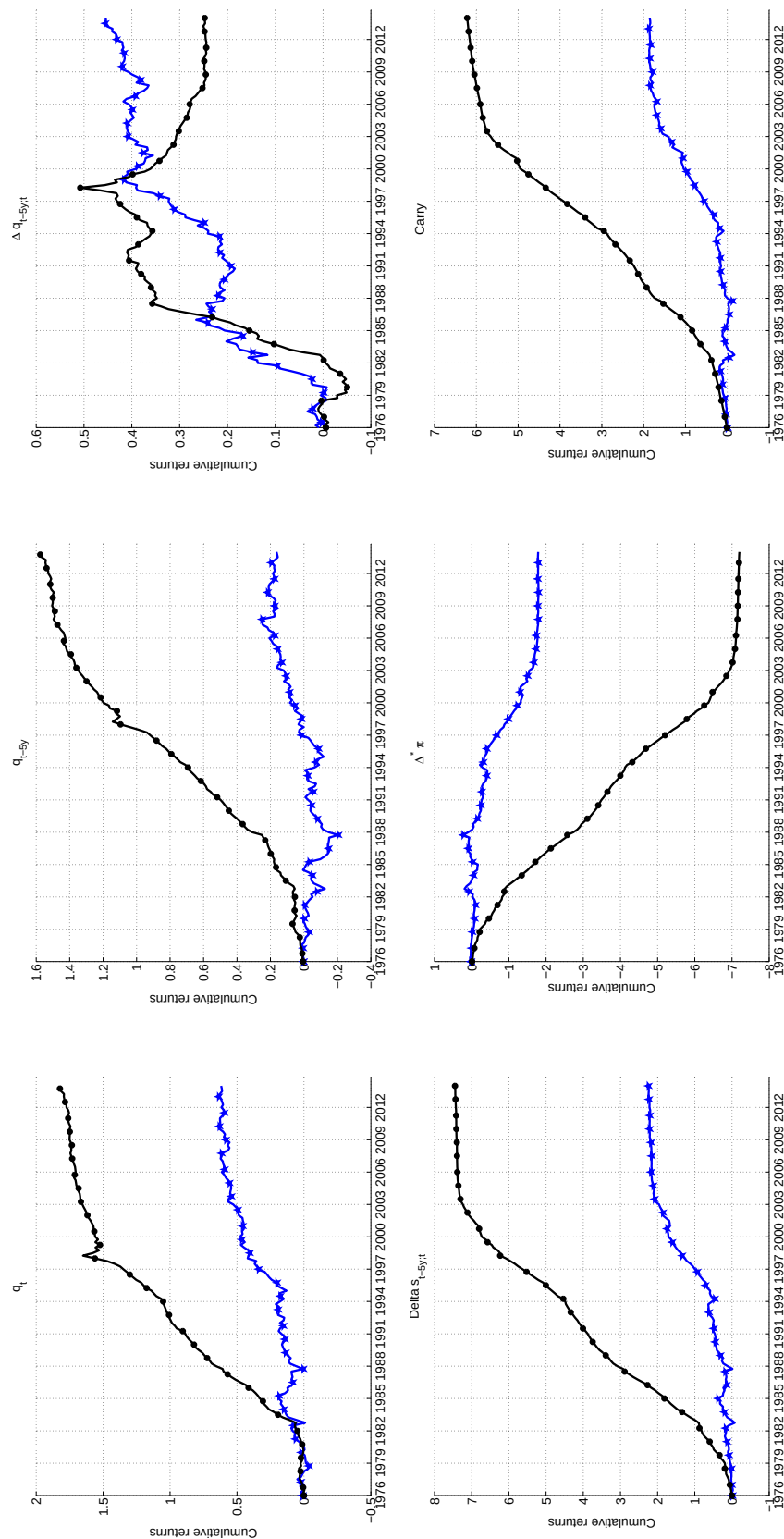
Panel A. Returns and exposures							
	q_t	$\Delta q_{t-5y,t}$	$-\Delta s_{t-5y,t}$	$-\Delta^* \pi_{t-5y,t}$	$\Delta q_{t-5y,t}^{HBS}$	$\Delta q_{t-5y,t}^{Qual}$	$\Delta q_{t-5y,t}^{RES}$
Unadjusted returns							
Mean	1.71	2.57	16.18	-13.61	-0.36	0.37	2.56
	[2.12]	[2.62]	[3.40]	[-3.08]	[-0.67]	[1.28]	[2.91]
<i>SR</i>	0.33	0.51	0.81	-0.71	-0.13	0.18	0.64
Exposure regressions							
α	0.18	3.12	7.15	-4.03	-0.16	0.19	3.09
	[0.26]	[3.18]	[2.77]	[-1.94]	[-0.27]	[0.53]	[3.25]
Carry	0.11	-0.40	1.27	-1.67	-0.10	-0.02	-0.28
	[1.17]	[-3.93]	[5.26]	[-8.82]	[-1.47]	[-0.61]	[-3.08]
Mom	-0.36	-0.58	-0.91	0.33	-0.12	-0.11	-0.35
	[-3.45]	[-4.52]	[-3.74]	[1.45]	[-1.45]	[-2.08]	[-3.05]
R^2	0.45	0.24	0.74	0.78	0.02	0.10	0.13
<i>IR</i>	0.05	0.71	0.70	-0.45	-0.06	0.09	0.83
Panel B. Global tangency portfolio							
	weights			t -stat			
Carry	0.32			[5.24]			
Momentum	0.26			[3.06]			
Value	0.07			[0.46]			
Fair value	0.35			[2.11]			

Table 10. Conditional and unconditional portfolios

This table reports average returns and Sharpe Ratios for unconditional and conditional portfolios. Unconditional portfolios are based on average values of signals whereas conditional portfolios are based on the most recent value of signals. The left part of Panel A shows results for the case where we build portfolios based on full-sample averages of the RER (q_t), the 5-year change in the real exchange rate ($\Delta q_{t-5y;t}$), and the fair value estimate based on adjusting RER for quality and HBS ($\Delta q_{t-5y;t}^{RES}$). The right part of Panel A shows results for conditional portfolios based on the same set of signals. The sample period here is 1976Q1 – 2014Q1 and the frequency is quarterly. The left part of Panel B uses average values of the signals from the first half of the sample (1976Q1 – 1995Q1) to compute portfolio weights that we apply for the second half of the sample period (1995Q2 – 2014Q1). The right part of Panel B shows results for conditional portfolios for the subsample 1995Q2 – 2014Q1 for comparison. t -statistics in brackets are based on Newey and West (1987) standard errors.

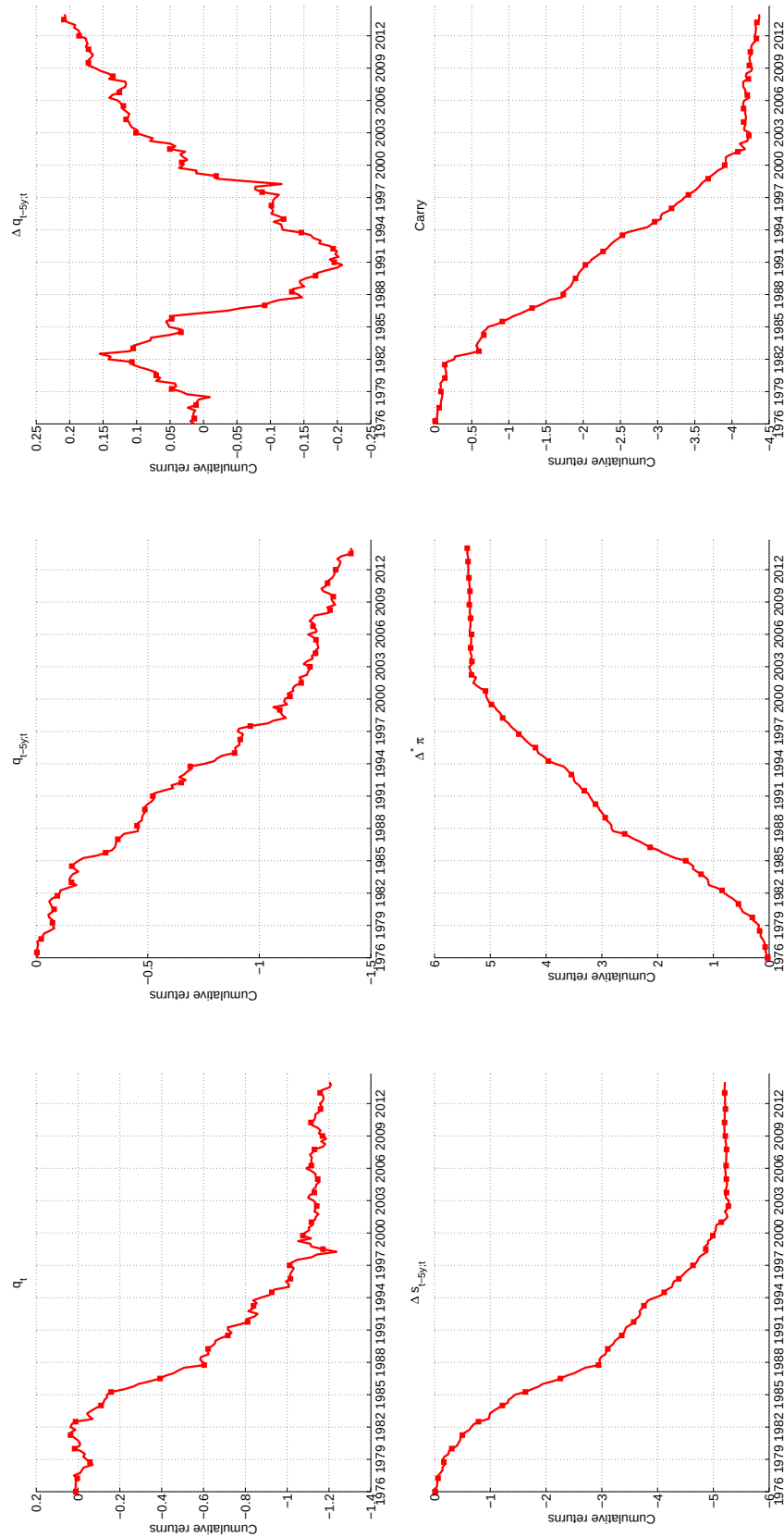
	Unconditional portfolios			Conditional portfolios		
	q_t	$\Delta q_{t-5y;t}$	$\Delta q_{t-5y;t}^{RES}$	q_t	$\Delta q_{t-5y;t}$	$\Delta q_{t-5y;t}^{RES}$
Panel A: Full sample averages						
<i>Excess returns</i>						
Mean	0.61	-0.08	-0.05	2.26	1.41	1.29
t	[0.78]	[-0.98]	[-0.97]	[2.74]	[2.82]	[2.90]
SR	0.13	-0.15	-0.18	0.42	0.51	0.62
<i>Exchange rate changes</i>						
Mean	-4.11	-0.30	-0.04	-3.62	0.49	1.37
t	[-4.55]	[-2.51]	[-1.01]	[-3.31]	[0.71]	[3.43]
Panel B: Averages based on first half of the sample (1976Q1 - 1995Q1)						
<i>Excess returns</i>						
Mean	3.47	0.99	0.22	3.62	1.25	1.22
t	[3.11]	[3.59]	[1.52]	[3.38]	[1.74]	[2.23]
SR	0.72	0.82	0.36	0.87	0.54	0.71
<i>Exchange rate changes</i>						
Mean	-2.68	-0.85	-0.15	-0.90	2.02	1.29
t	[-2.39]	[-2.31]	[-1.17]	[-0.66]	[3.04]	[2.24]

Figure 1. Cumulative returns: Excess returns and carry



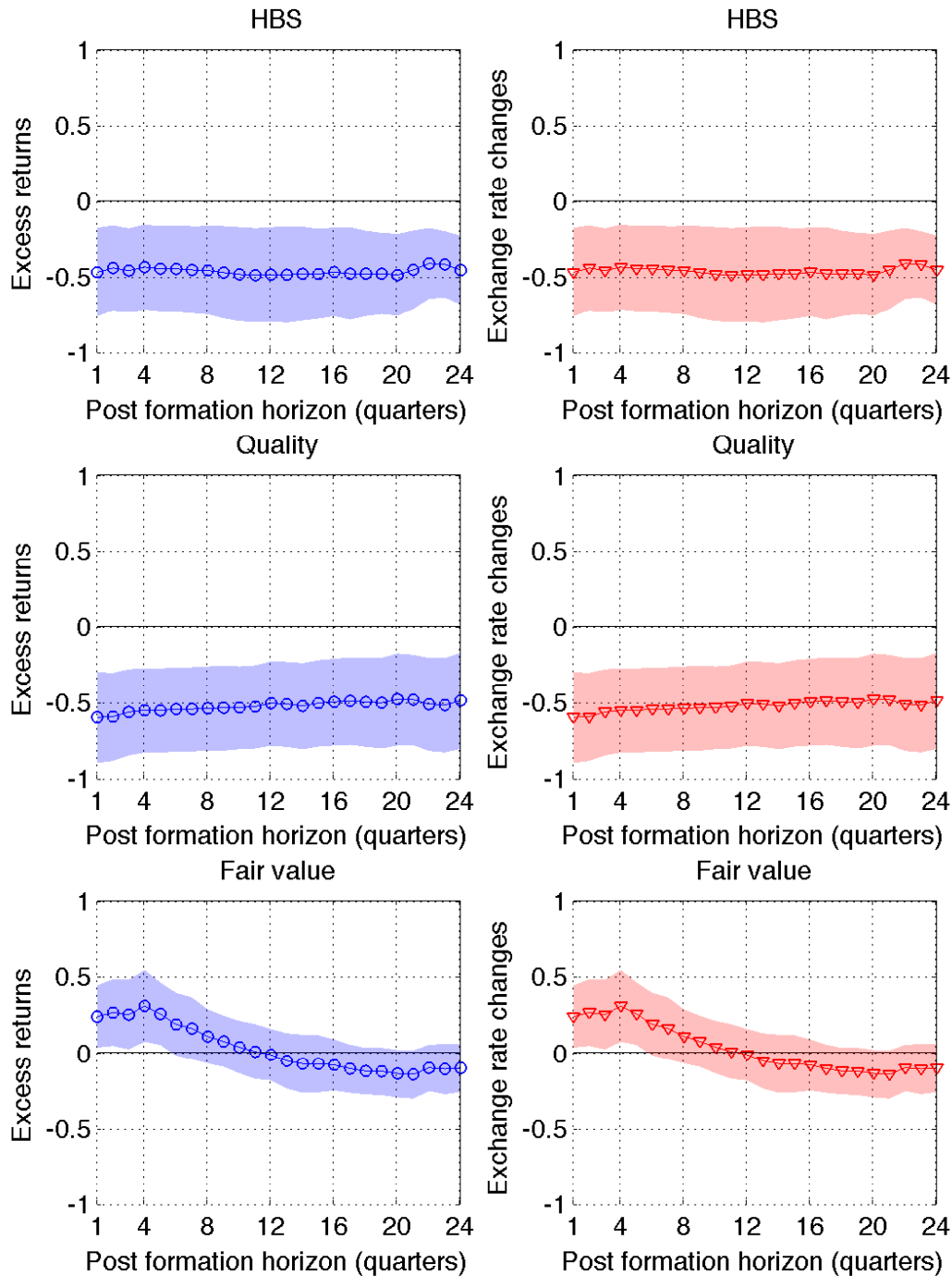
This figure plots cumulative excess returns (blue, stars) and the carry component of excess returns (black, circles) for different portfolios relating to real exchange rates. We also plot the same for a carry trade portfolio for comparison.

Figure 2. Cumulative returns: Spot exchange rate returns



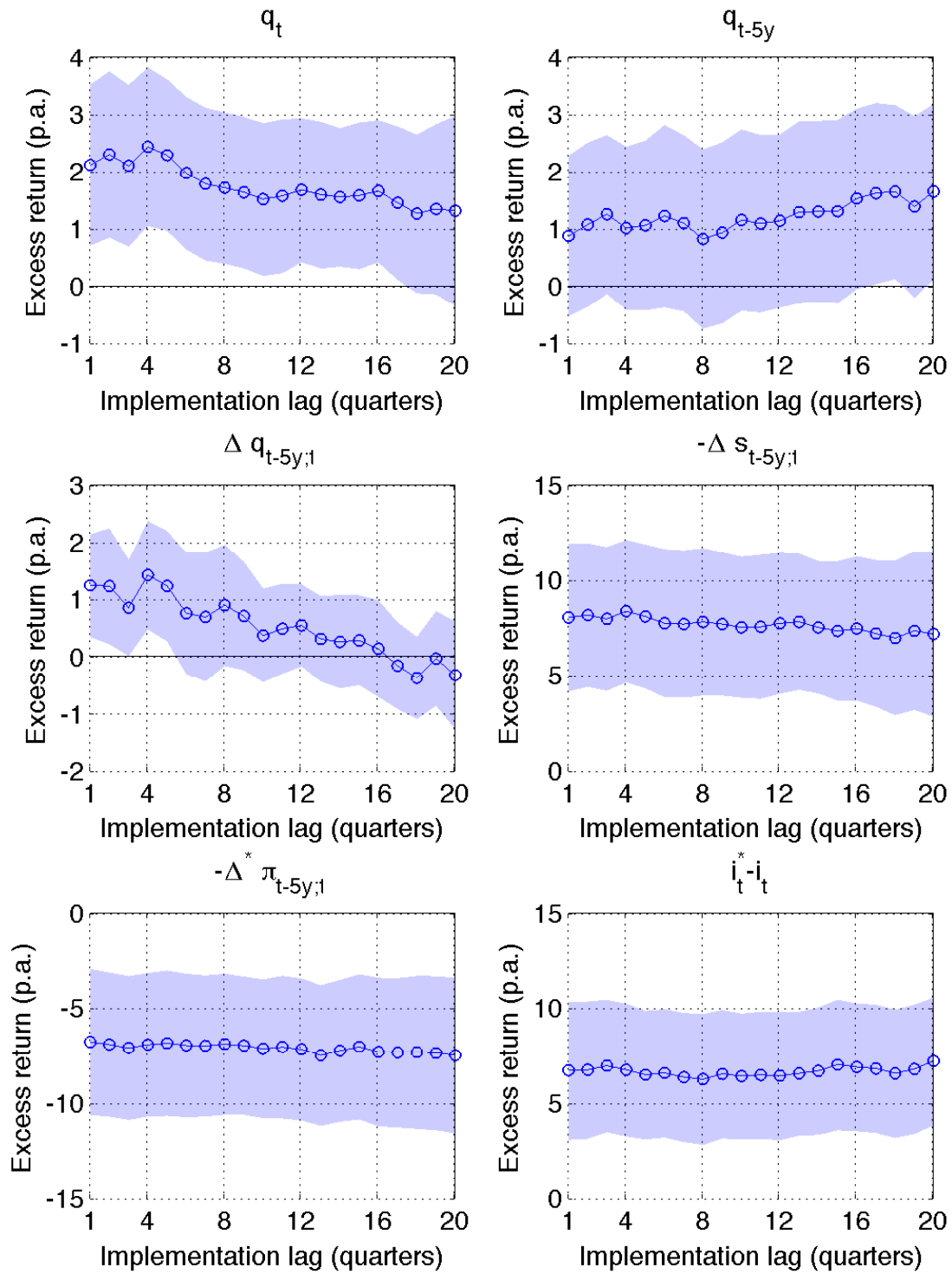
This figure plots cumulative spot exchange rate changes (red, squares) for different portfolios relating to real exchange rates. We also plot the same for a carry trade portfolio for comparison.

Figure 3. Fair value portfolios: Post-formation return drift



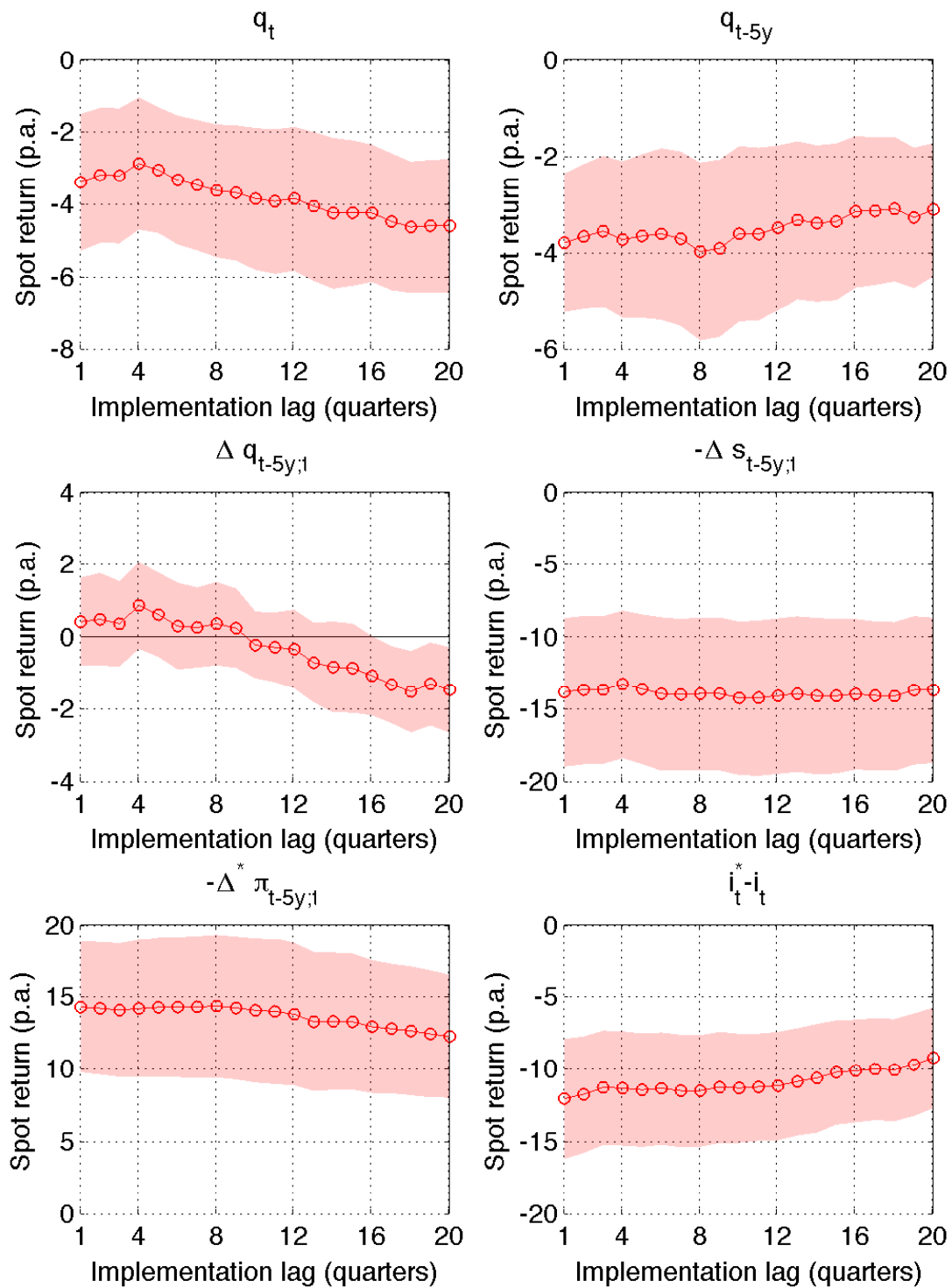
This figure plots post-formation portfolio excess returns and exchange rate changes for HBS, quality, and fair value portfolios, where the latter are obtained from cross-sectionally regressing real exchange rates on real per capita GDP growth (HBS effect) and export quality and using the residuals as signals to form portfolios. The figure shows returns for 1, 2, ..., 24 quarters after portfolio formation.

Figure 4. Implementation lags: Excess returns



This figure plots average annualized excess returns for rank portfolios based on different signals and for implementation lags (lags between observing the signal and the formation of the portfolio).

Figure 5. Implementation lags: Spot exchange rate changes



This figure plots average annualized spot exchange rate changes for rank portfolios based on different signals and for implementation lags (lags between observing the signal and the formation of the portfolio).

Supplementary Internet Appendix to accompany
Currency Value

Table A.I. Return correlations

This table shows correlation coefficients of portfolio returns based on different signals, where q_t denotes real exchange rates, q_{t-5y} denotes lagged real exchange rates (five-year lag), $\Delta q_{t-5y;t}$ denotes five-year changes in real exchange rates, $-\Delta s_{t-5y;t}$ denotes negative five-year nominal spot exchange rate changes (depreciation rates of foreign countries), $-\Delta^* \pi_{t-5y;t}$ denotes the negative of five-year inflation differentials (i.e. inflation differentials between the U.S. and foreign countries), and $\Delta^* i_t$ denotes (nominal) short-term interest rate differentials (foreign country minus U.S.). Real exchange rates are defined such that a higher real exchange rate indicates a weaker foreign currency. Hence, the portfolios go long currencies with a relatively low valuation and go short currencies with a relatively high valuation. Excess returns and exchange rates are defined such that positive numbers mean a positive return on holding the foreign currency. The sample period is 1976Q1 – 2014Q1 and the frequency is quarterly.

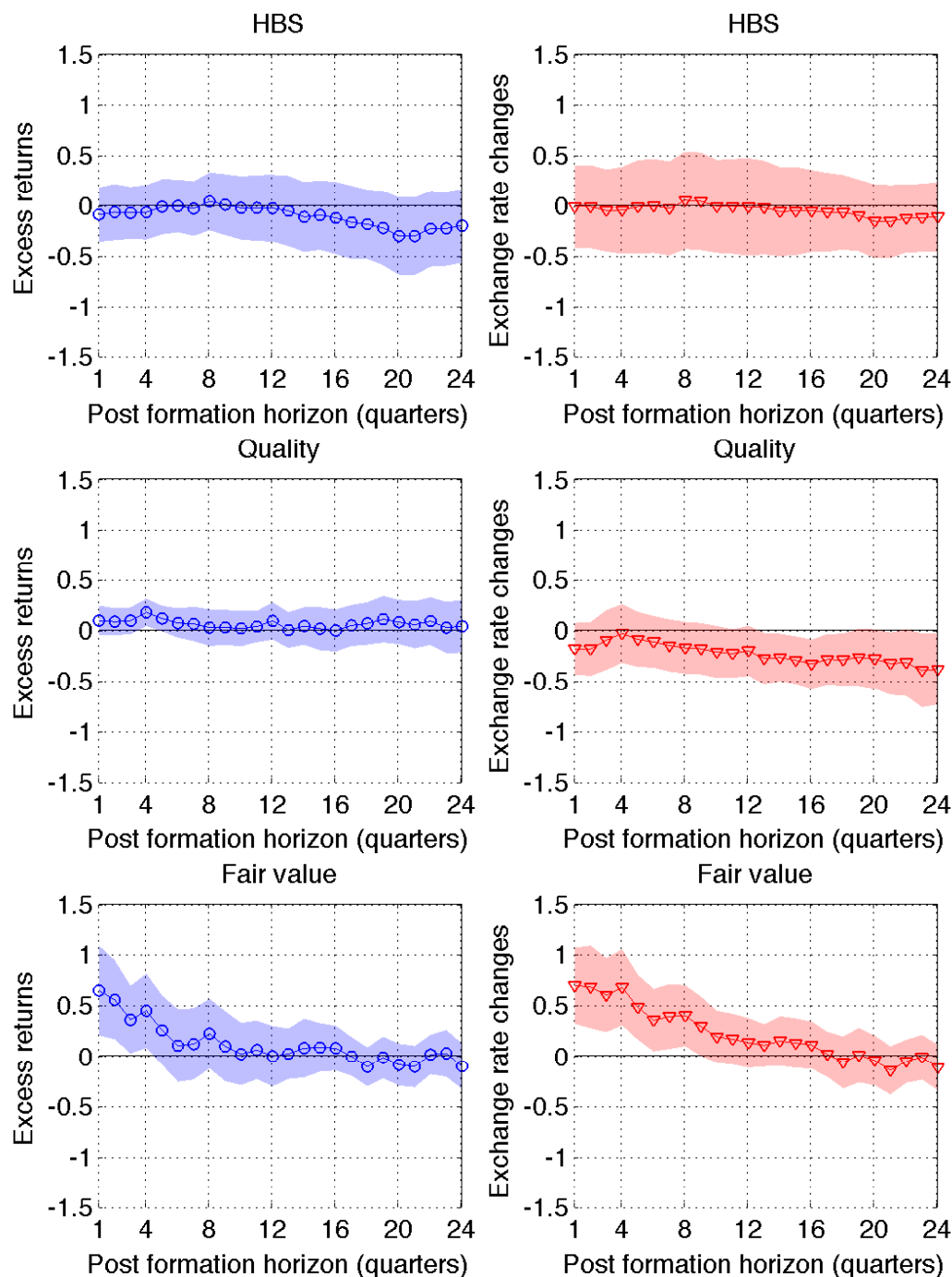
	q_t	q_{t-5y}	$\Delta q_{t-5y;t}$	$-\Delta s_{t-5y;t}$	$-\Delta^* \pi_{t-5y;t}$	$\Delta^* i_t$
Panel A. Excess returns						
q_t	1.00	0.84	0.45	0.73	-0.64	0.60
q_{t-5y}	0.84	1.00	-0.10	0.61	-0.67	0.67
$\Delta q_{t-5y;t}$	0.45	-0.10	1.00	0.33	-0.06	-0.00
$\Delta s_{t-5y;t}$	0.73	0.61	0.33	1.00	-0.96	0.85
$\Delta^* \pi_{t-5y;t}$	-0.64	-0.67	-0.06	-0.96	1.00	-0.90
$\Delta^* i_t$	0.60	0.67	-0.00	0.85	-0.90	1.00
Panel B. Spot exchange rate changes						
q_t	1.00	0.83	0.61	0.67	-0.54	0.48
q_{t-5y}	0.83	1.00	0.05	0.48	-0.52	0.55
$\Delta q_{t-5y;t}$	0.61	0.05	1.00	0.51	-0.22	0.06
$\Delta s_{t-5y;t}$	0.67	0.48	0.51	1.00	-0.95	0.77
$\Delta^* \pi_{t-5y;t}$	-0.54	-0.52	-0.22	-0.95	1.00	-0.85
$\Delta^* i_t$	0.48	0.55	0.06	0.77	-0.85	1.00

Table A.II. Portfolio returns: Influential currencies?

This table reports returns to portfolio based on linear weights and using RER levels (left part of the table) or 5-year changes in RERs (right part) where we eliminate one country at a time from the investment universe. We report both average excess returns and average exchange rate changes. Average returns and spot rate changes are annualized and SR denotes the (annualized) Sharpe Ratio. t -statistics in brackets are based on Newey and West (1987) standard errors. The sample period is 1976Q1 – 2014Q1 and the frequency is quarterly.

	RER levels				5-year RER changes			
	Excess returns		Exchange rate changes		Excess returns		Exchange rate changes	
	mean	t	SR	mean	t	mean	t	mean
Australia	1.59	[2.31]	0.34	-3.24	[-3.38]	1.12	[2.62]	0.46
Canada	1.70	[2.47]	0.36	-3.38	[-3.48]	1.21	[2.76]	0.48
France	1.60	[2.33]	0.34	-3.17	[-3.33]	1.16	[2.67]	0.47
Germany	1.60	[2.36]	0.35	-3.11	[-3.28]	1.18	[2.69]	0.47
Hungary	1.71	[2.49]	0.37	-3.40	[-3.61]	1.18	[2.67]	0.47
India	1.85	[2.48]	0.38	-3.24	[-3.17]	1.28	[2.71]	0.48
Indonesia	1.69	[2.49]	0.37	-3.26	[-3.67]	1.17	[2.69]	0.47
Italy	1.59	[2.28]	0.34	-3.18	[-3.31]	1.23	[2.75]	0.49
Japan	1.57	[2.29]	0.34	-2.82	[-2.89]	1.26	[2.49]	0.45
Korea	1.44	[2.11]	0.31	-3.36	[-3.41]	1.02	[2.33]	0.41
Mexico	1.26	[2.19]	0.34	-2.56	[-3.25]	0.96	[2.60]	0.44
New Z.	1.67	[2.45]	0.36	-3.19	[-3.27]	1.20	[2.77]	0.49
Norway	1.61	[2.31]	0.34	-3.18	[-3.29]	1.19	[2.63]	0.46
Singapore	1.62	[2.33]	0.34	-3.17	[-3.29]	1.23	[2.89]	0.50
S. Africa	1.82	[2.55]	0.38	-3.25	[-3.22]	1.12	[2.43]	0.42
Spain	1.72	[2.47]	0.36	-3.30	[-3.43]	1.22	[2.75]	0.47
Sweden	1.65	[2.31]	0.35	-3.27	[-3.41]	1.20	[2.68]	0.47
Switzerland	1.59	[2.46]	0.36	-2.80	[-2.95]	1.29	[2.68]	0.48
Taiwan	1.55	[2.14]	0.31	-3.20	[-3.26]	1.27	[2.81]	0.49
Thailand	1.72	[2.48]	0.37	-3.35	[-3.45]	1.20	[2.81]	0.48
Turkey	1.28	[1.92]	0.27	-2.40	[-2.71]	1.22	[2.91]	0.50
U.K.	1.61	[2.29]	0.34	-3.20	[-3.33]	1.20	[2.67]	0.47

Figure A.1. Fair value portfolios based on 5-year changes in RER: Post-formation return drift



This figure plots post-formation portfolio excess returns and exchange rate changes for HBS, quality, and fair value portfolios, where the latter are obtained from cross-sectionally regressing 5-year real exchange rate changes on real per capita GDP growth (HBS effect) and export quality and using the residuals as signals to form portfolios. The figure shows returns for 1, 2, ..., 24 quarters after portfolio formation.